Schlumberger

SonicScope - BestDT*
Multipole DT Computation

Techlog Acoustics Interpretation Suite

Company: Equinor

Well: 31/5-7

Field: Eos

Rig Name: West Hercules

Country: Norway

Latitude:	60° 34' 35.14" N	UWID:	
Longitude:	3° 26' 36.13" E	Rig Name:	West Hercules
Range:		Rig Type:	Semi-Submersible

FL: Norwegian North Sea
FL1: Section: 8 1/2 in. pilot
FL2: Job no.: 19SCA0085

Log Measured From - Drill Floor: 31 m

Permanent Datum - 0 m

Ground Level: -307 m

Acquisition Dates:	4-Dec-2019 -- 4-Dec-2019		
Run number - Print Interval:	Run2 -- 389m to 916m		
Index Type:	Measured Depth		
Index Scale:	1:200		
Depth Source:	Drillers depth		
Mud Type:	Seawater		
Conveyance:	Drill Pipe		
Print Type:	SIS		
Processing Date:	7-Dec-2019	Log Analyst: TK	Center: Stavanger
D&M ref:	19SCA0085		Techlog 2017.2

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Disclaimer

Executive Summary

Monopole and quadrupole sonic waveforms were recorded by the SonicScope tool in the 8.5in section of the well 31/5-7.

Drill summary

SonicScope data was acquired in the 8.5in borehole which was drilled from 411.5m to 944m as Run 2. The 30in casing shoe was at 406.6m.

Monopole:

The monopole waveform data was processed to obtain formation compressional slowness.

Compressional slowness was continuous in the monopole waveform over the range 119 us/ft to 189 us/ft.

Compressional slownesses were achieved using Leaky-P processing.

Quadrupole:

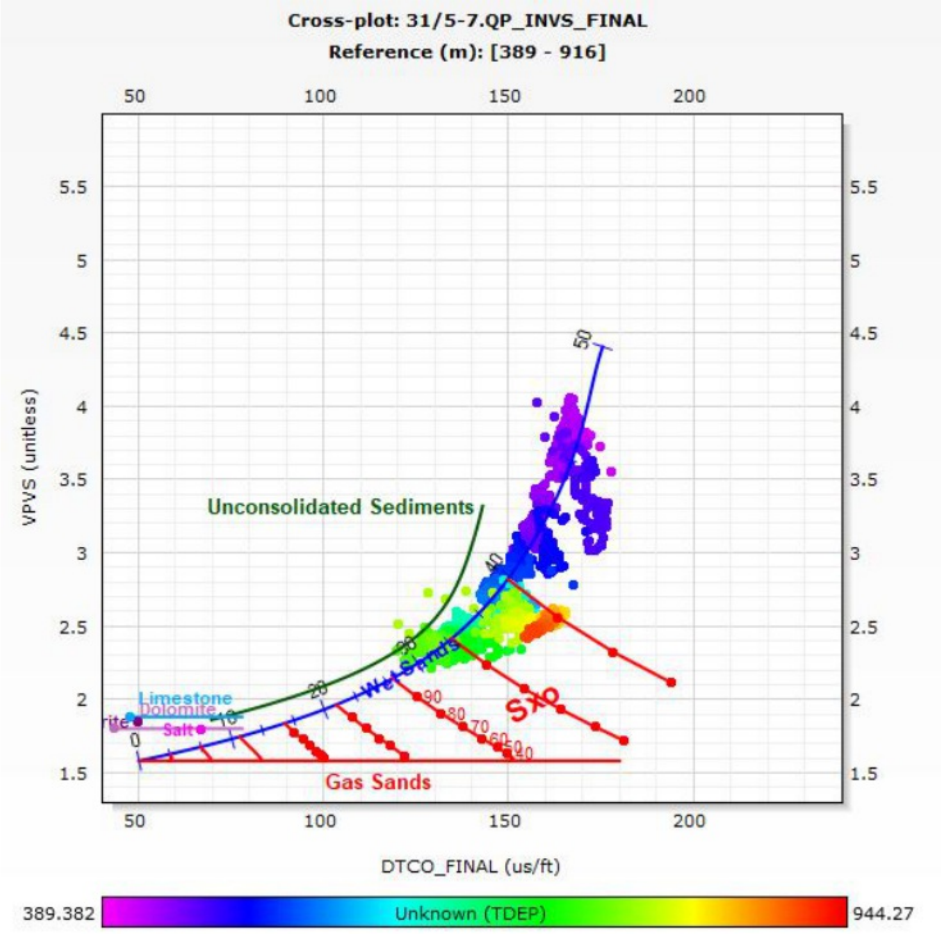
The quadrupole waveform data was processed to obtain formation shear slowness covering the range 275 us/ft to 677 us/ft.

Intermittent quadrupole shear slowness was computed in the entire interval.

Quadrupole shear was used as final shear slowness.

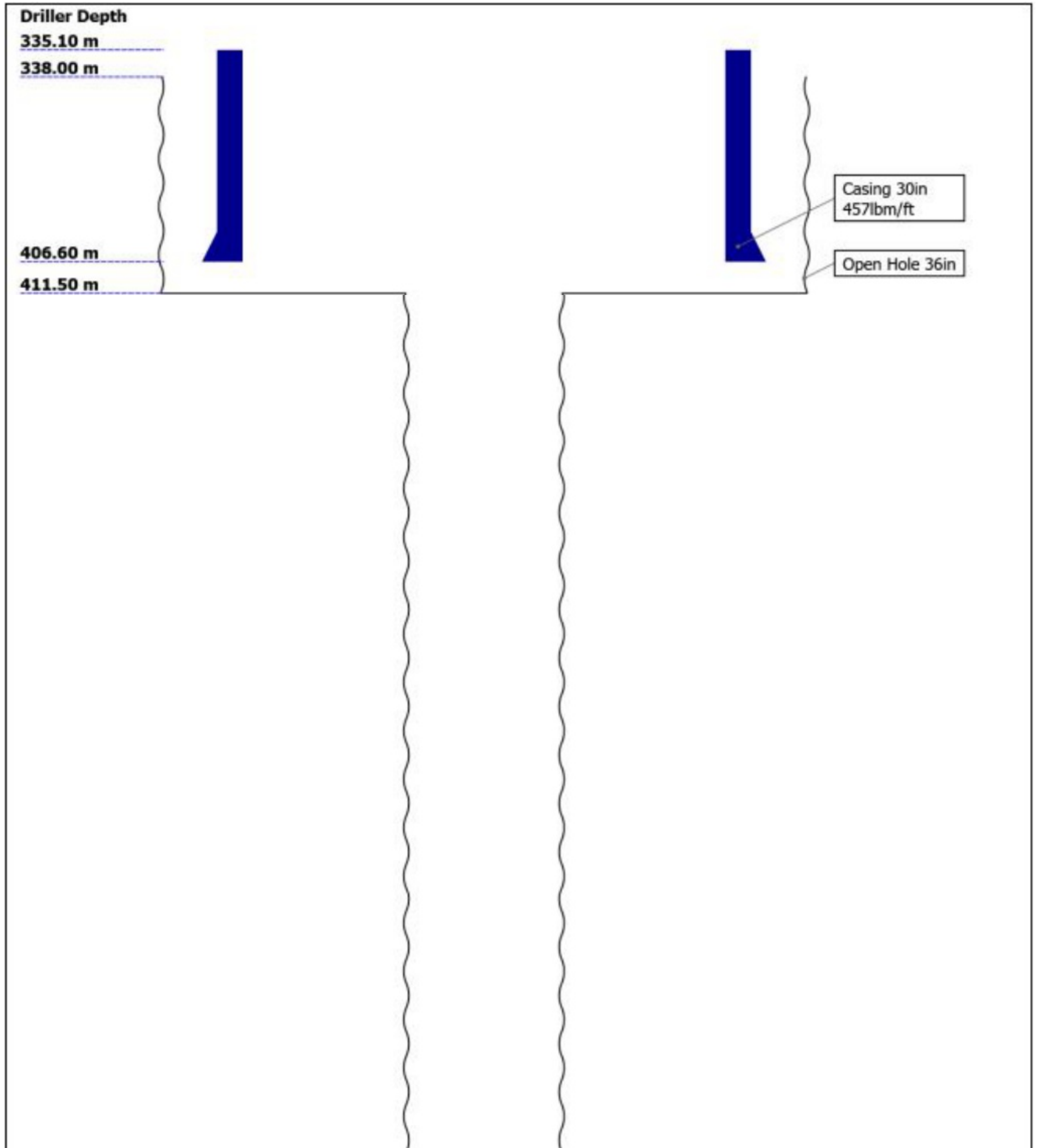
Results:

The analysis and the processing of the monopole and quadrupole waveform data confirmed the quality of the SonicScope waveforms.

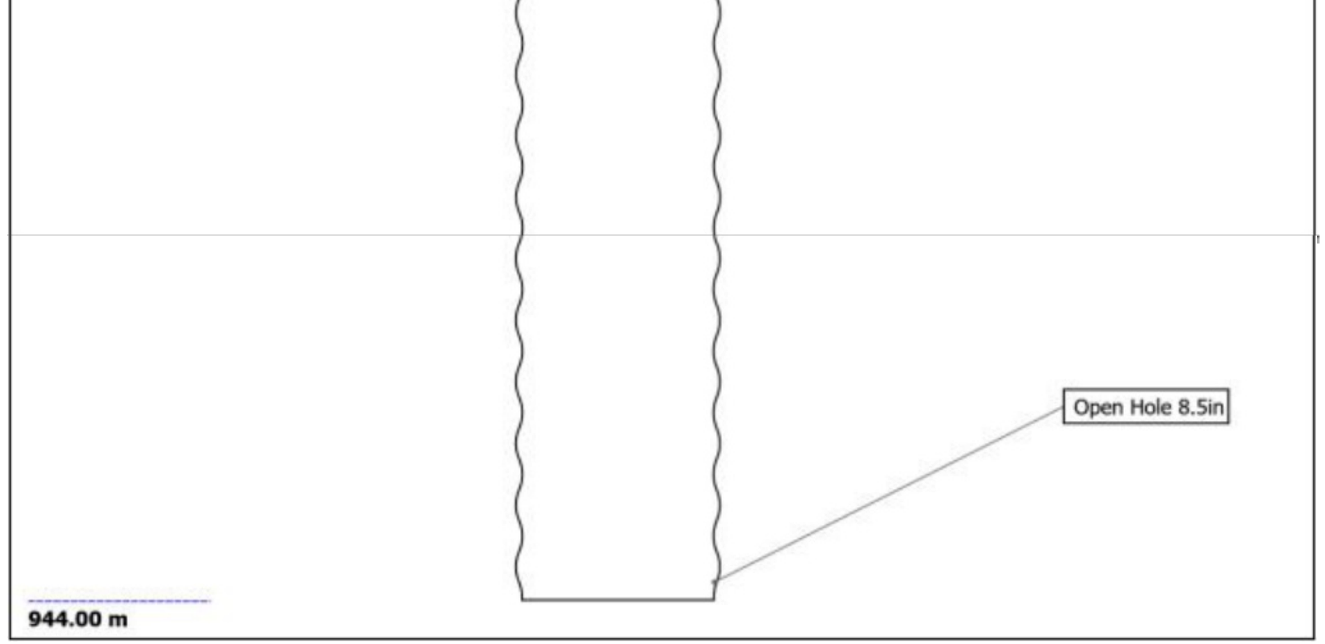


VPVS crossplot

Well Sketch



Driller Depth
335.10 m
338.00 m
406.60 m
411.50 m
Casing 30in
457lbm/ft
Open Hole 36in


Open Hole 8.5in
944.00 m

Remarks & Equipment Summary

Run 2: Toolstring

Equip name	Length		MP name	Offset
SONICSCOPE6:H 0121 MP6DC:H0121 Upper Stabilizer:00 164-01 Lower Stabilizer:00 370-2-2 Upper Extender MP6UC:8019 MP6LC:8019 Lower Extender:RF E-11 XOS6:78001	32.32		SonicScope ROP Delta-T	 29.33 28.12
TELE675:E4699 MSSU675:12775-9 Upper Extender:08-118 MDC675:E4699 MMA:3766 MDI:1277 PMVC:560 PMEA:3491 MTA:2375 MTK675:1510/2989	22.34		TeleScope	

Run 2: Remarks

All depths are referenced to driller's depth. See EOWR for depth tracking details.

All data from tool memory. All data acquired while drilling.

Gamma Ray measurement is environmentally corrected for mud weight, bit size, collar thickness and neutron activation.

Resistivity measurements are borehole compensated and environmentally corrected for bit size, mud resistivity and temperature.

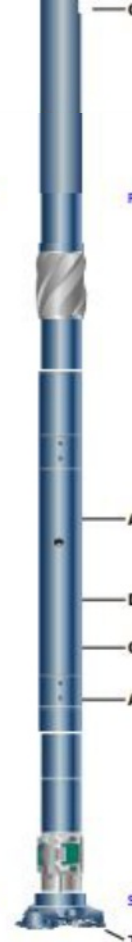
Bulk Density is compensated for tool standoff/mud cake.

Neutron Porosity is calculated with limestone matrix and is environmentally corrected for bit size, mud weight, temperature, pressure and mud salinity.

Run 2 and 8 1/2 in. pilot section TD at 944 m.

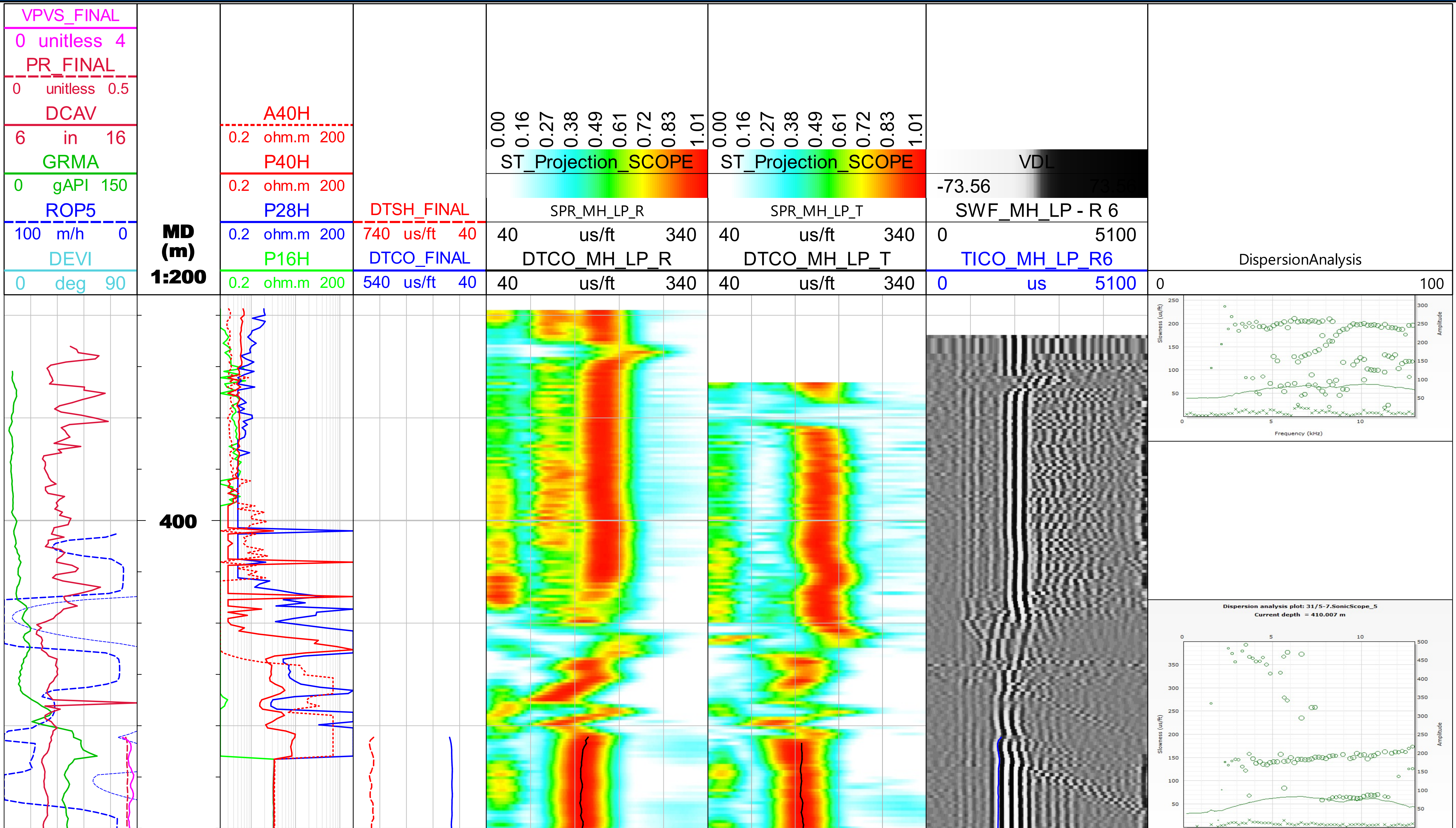
15
MSSD675:916772-1
Lower Extender, Long:319

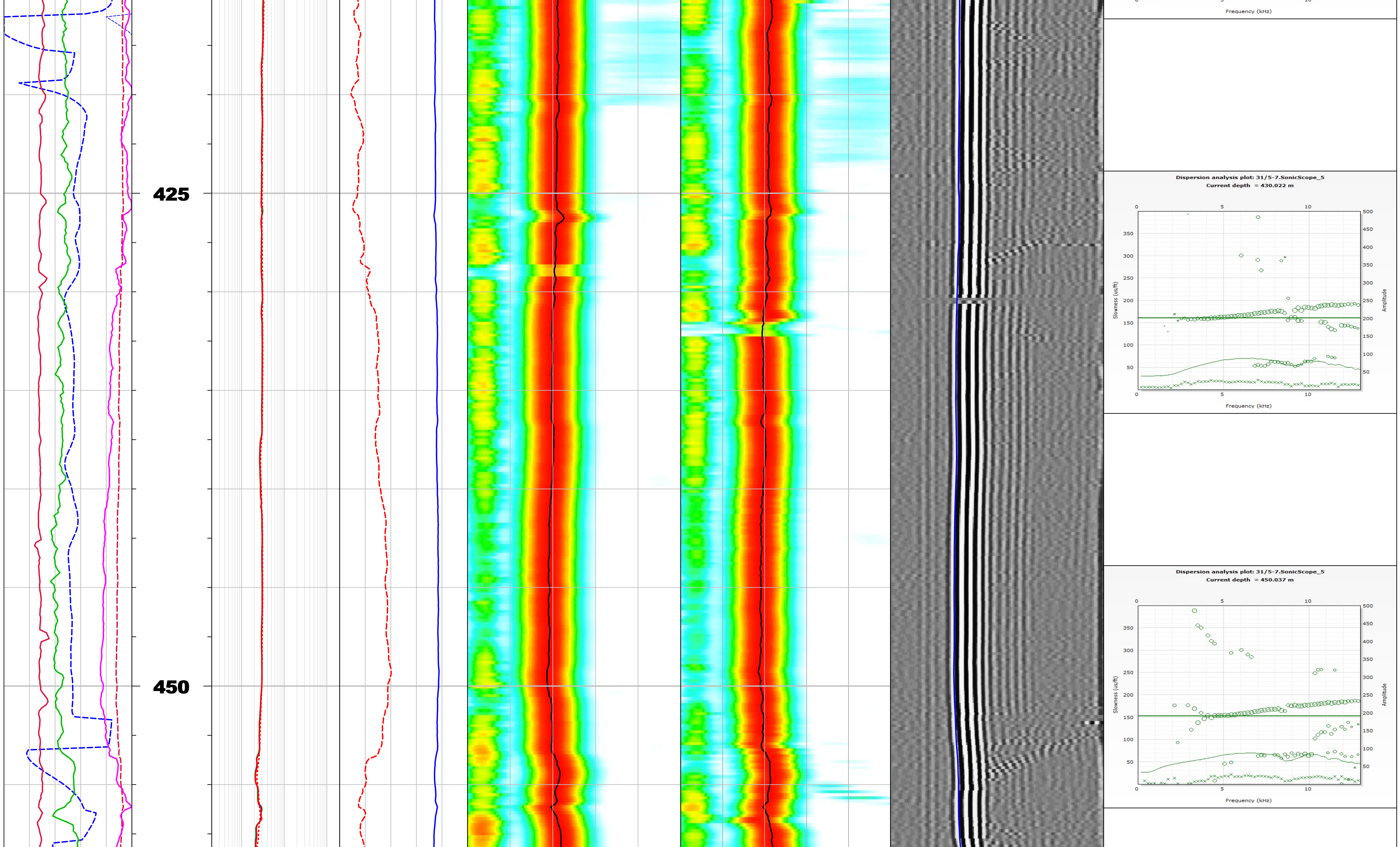
DV6MT:2855 **13.96**
DVMC:2855
DVMS:3811
UpperExtender:1954
PNG:8718-51675
DVME:1801
GGLS:A2609
LowerExtender:08/044
MSSA:83777



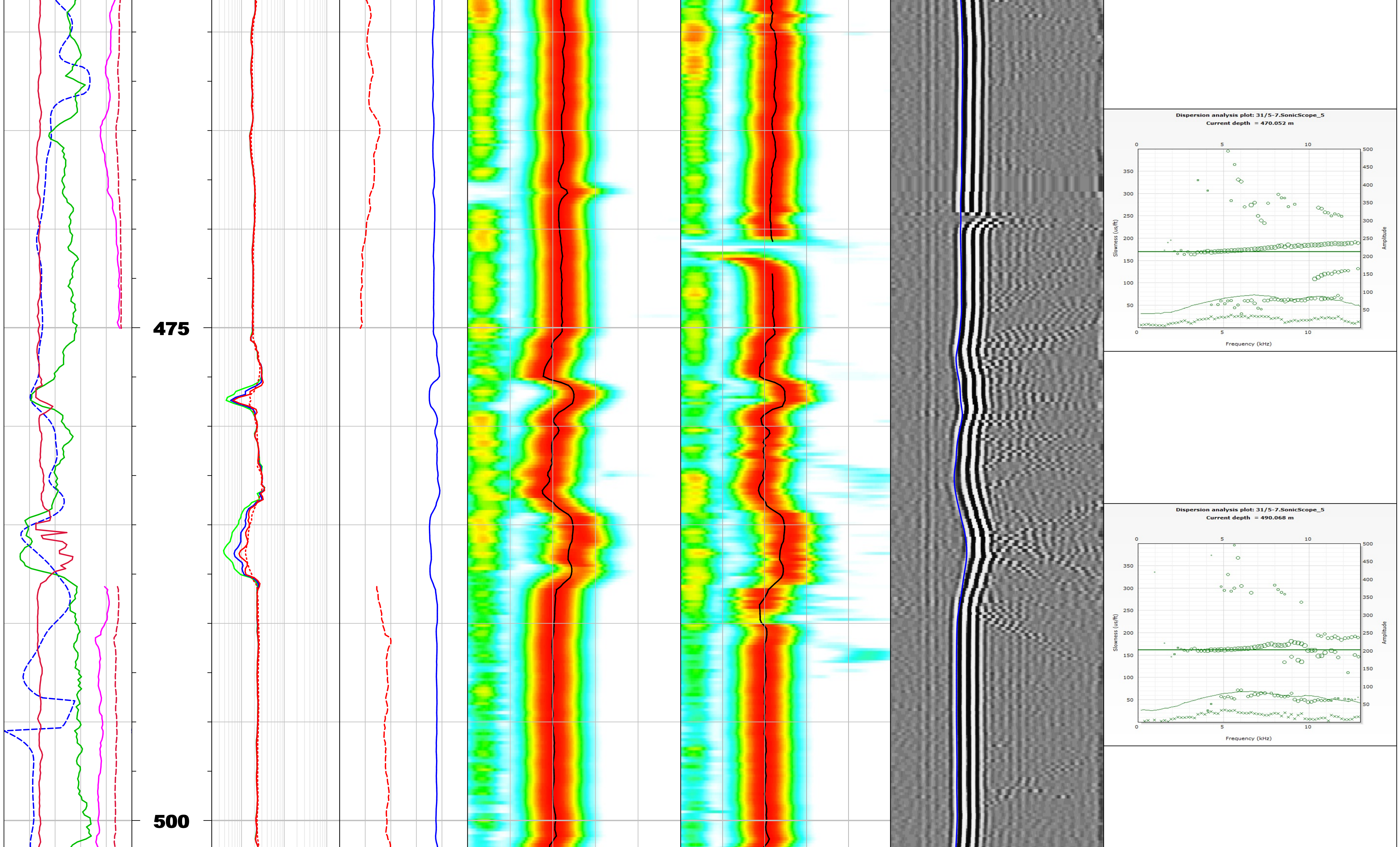
Lengths are in m
Maximum Outer Diameter = 8.500 in
Line: Sensor Location, Value: Gating Offset
All measurements are relative to TOOL_ZERO

DT Monopole High Log





Frequency (kHz)
425
450
Dispersion analysis plot: 31/5-7.SonicScope_5
Current depth = 430.022 m
Slowness (us/ft)
Amplitude
Frequency (kHz)
Dispersion analysis plot: 31/5-7.SonicScope_5
Current depth = 450.037 m
Slowness (us/ft)
Amplitude
Frequency (kHz)



475
500
Dispersion analysis plot: 31/5-7.SonicScope_5
Current depth = 470.052 m
Slowness (us/ft)
Amplitude
Frequency (kHz)
Dispersion analysis plot: 31/5-7.SonicScope_5
Current depth = 490.068 m
Slowness (us/ft)
Amplitude
Frequency (kHz)


525
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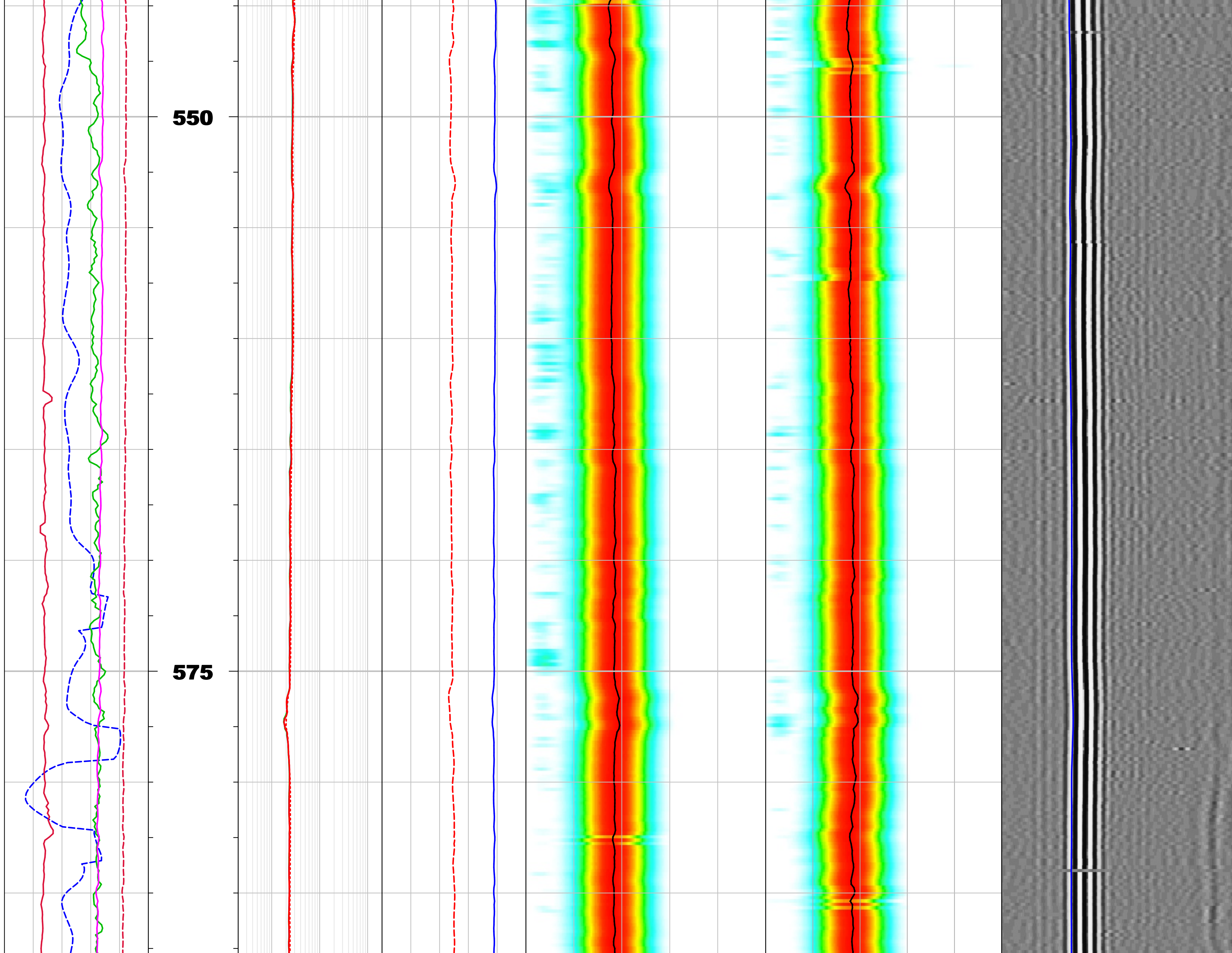
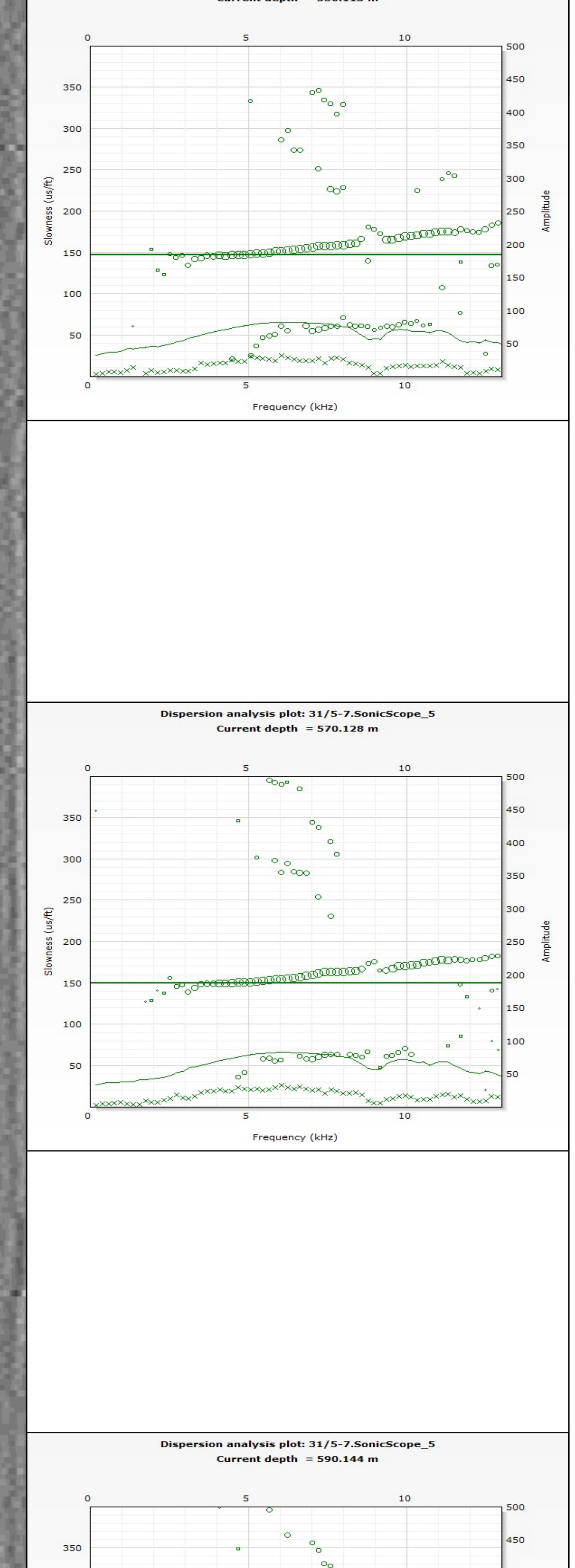

550
575


Slowness (us/ft)
Amplitude
Frequency (kHz)
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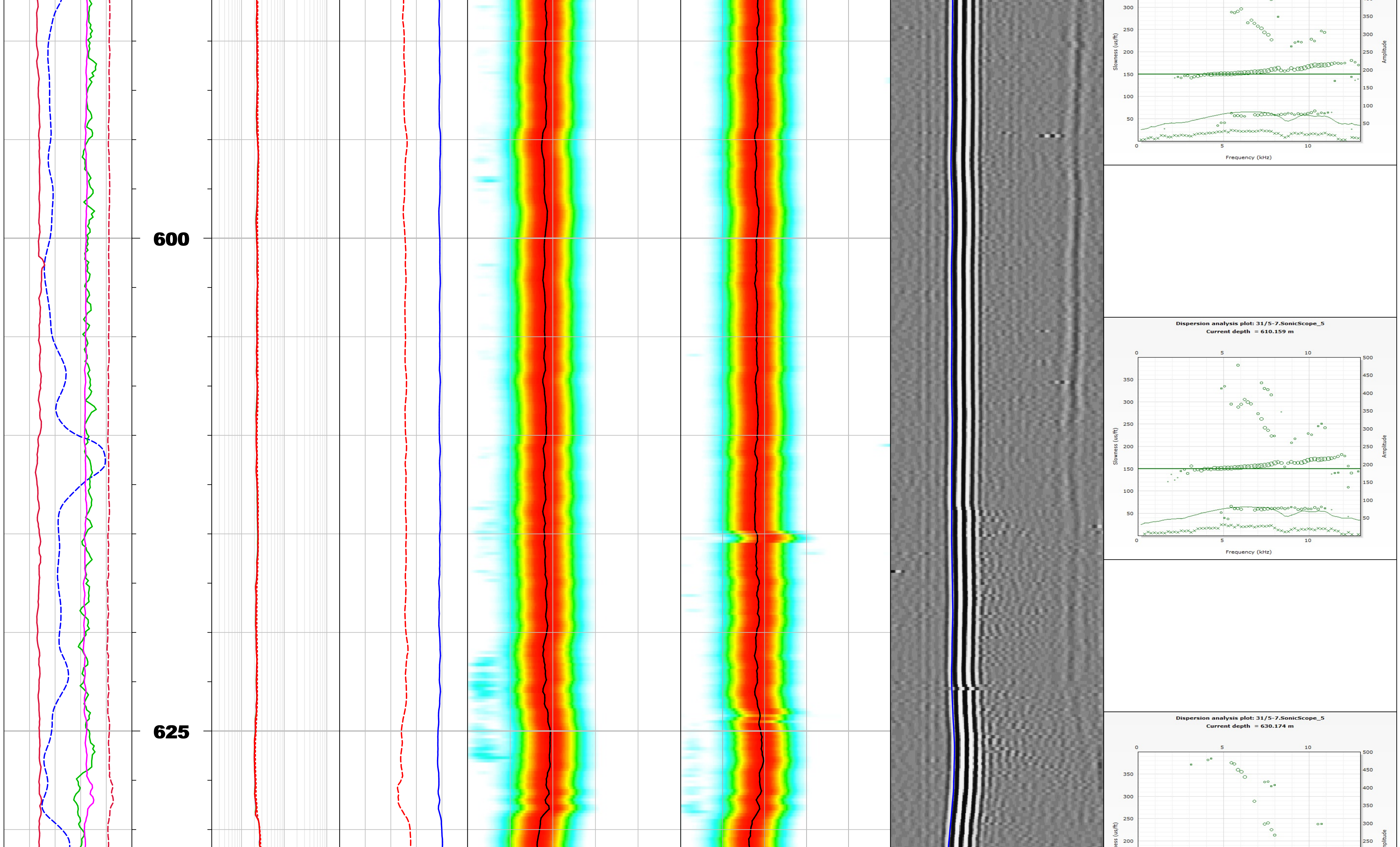
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650
Frequency (kHz)
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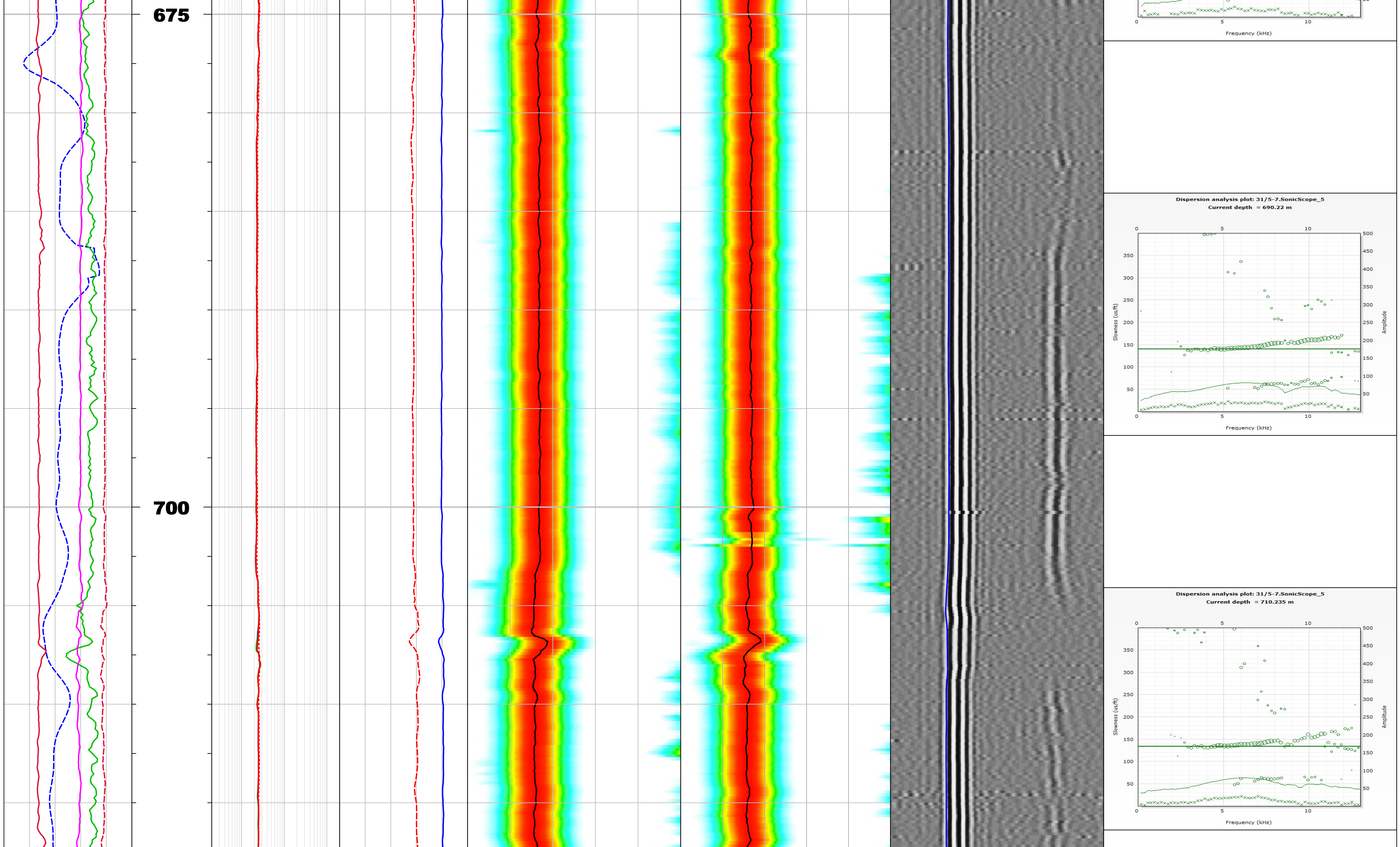
675
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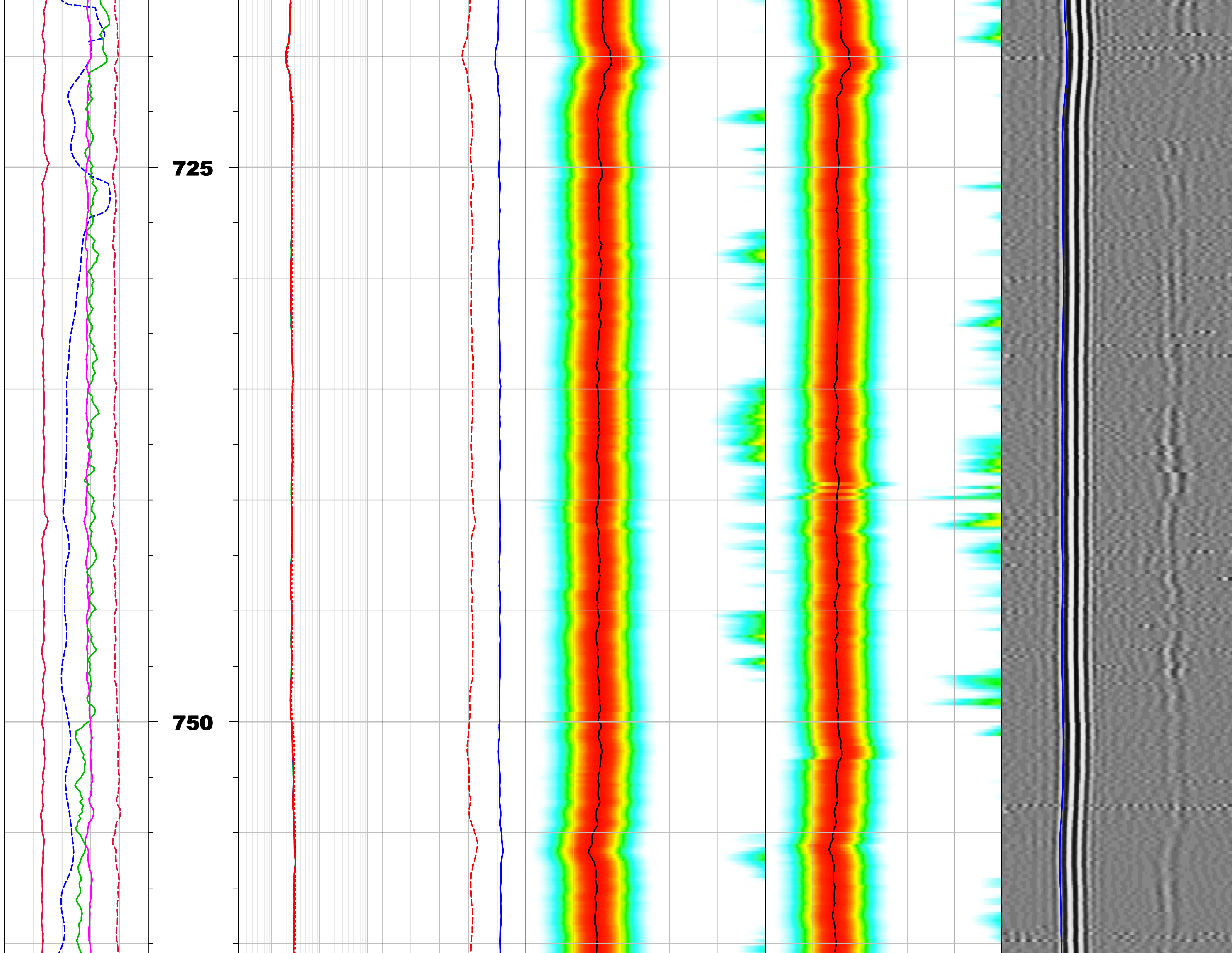
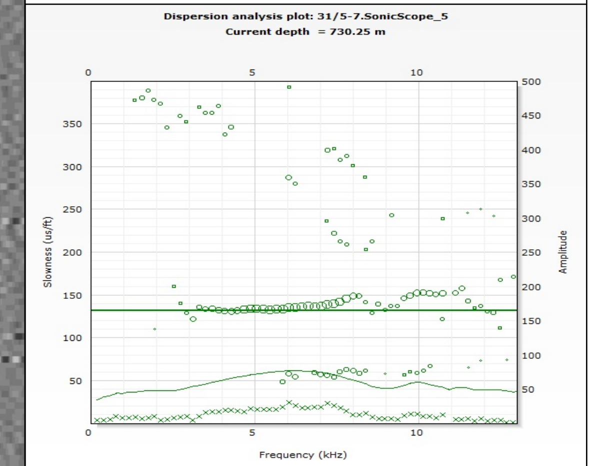

725
750


Dispersion analysis plot: 31/5-7.SonicScope_5
Current depth = 730.25 m
Slowness (us/ft)
Amplitude
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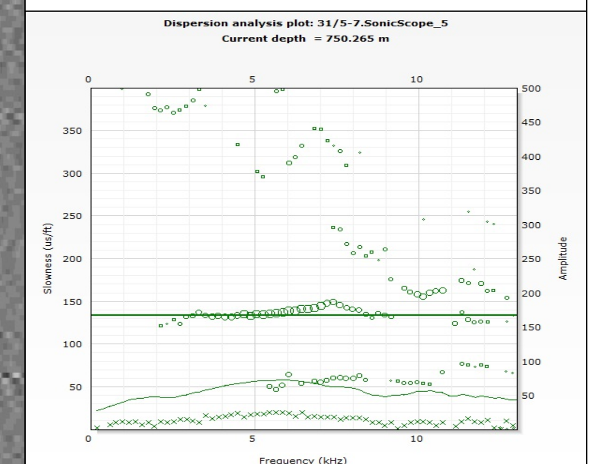
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Current depth = 750.265 m
Slowness (us/ft)
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775
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Current depth = 850.341 m


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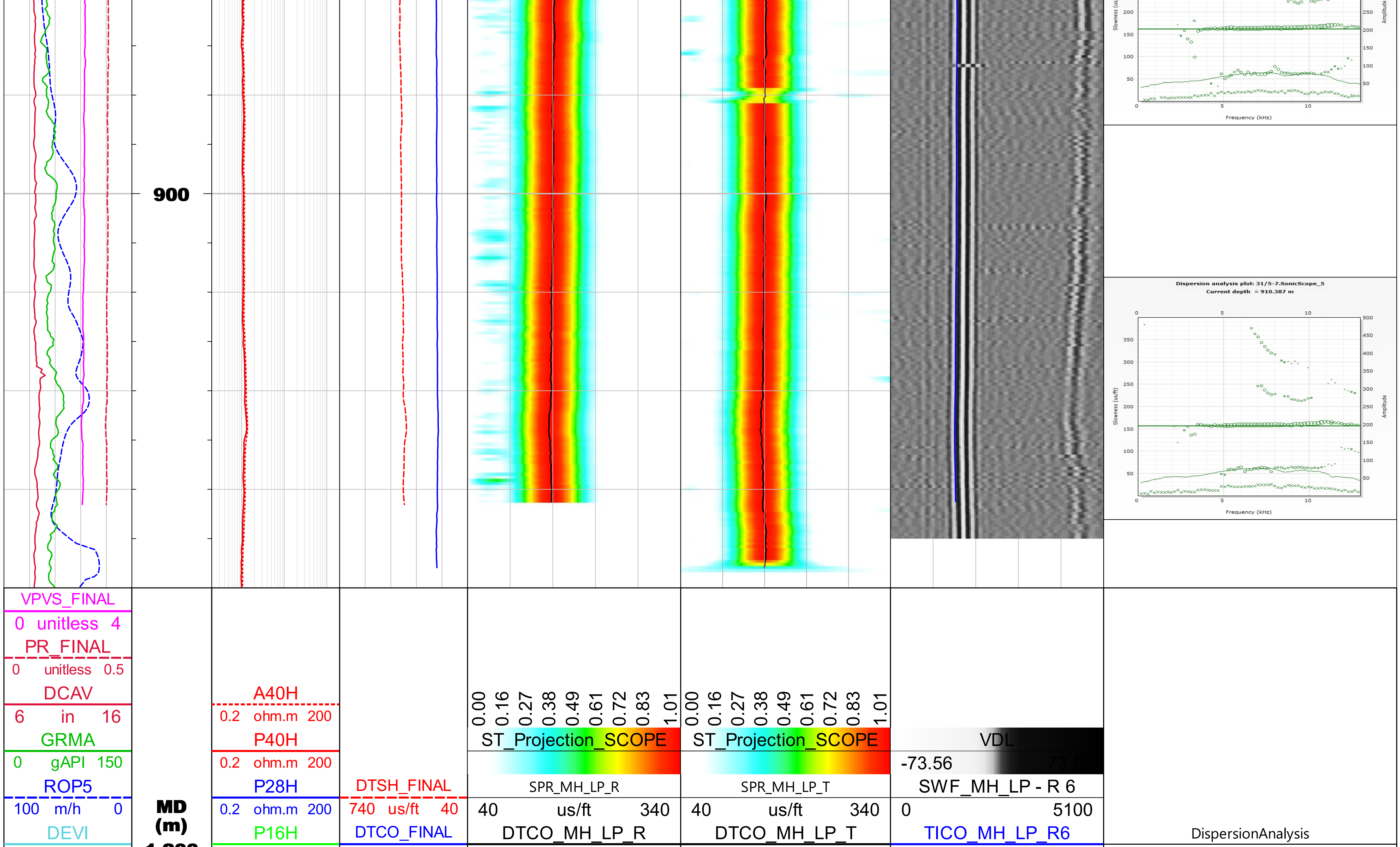

900
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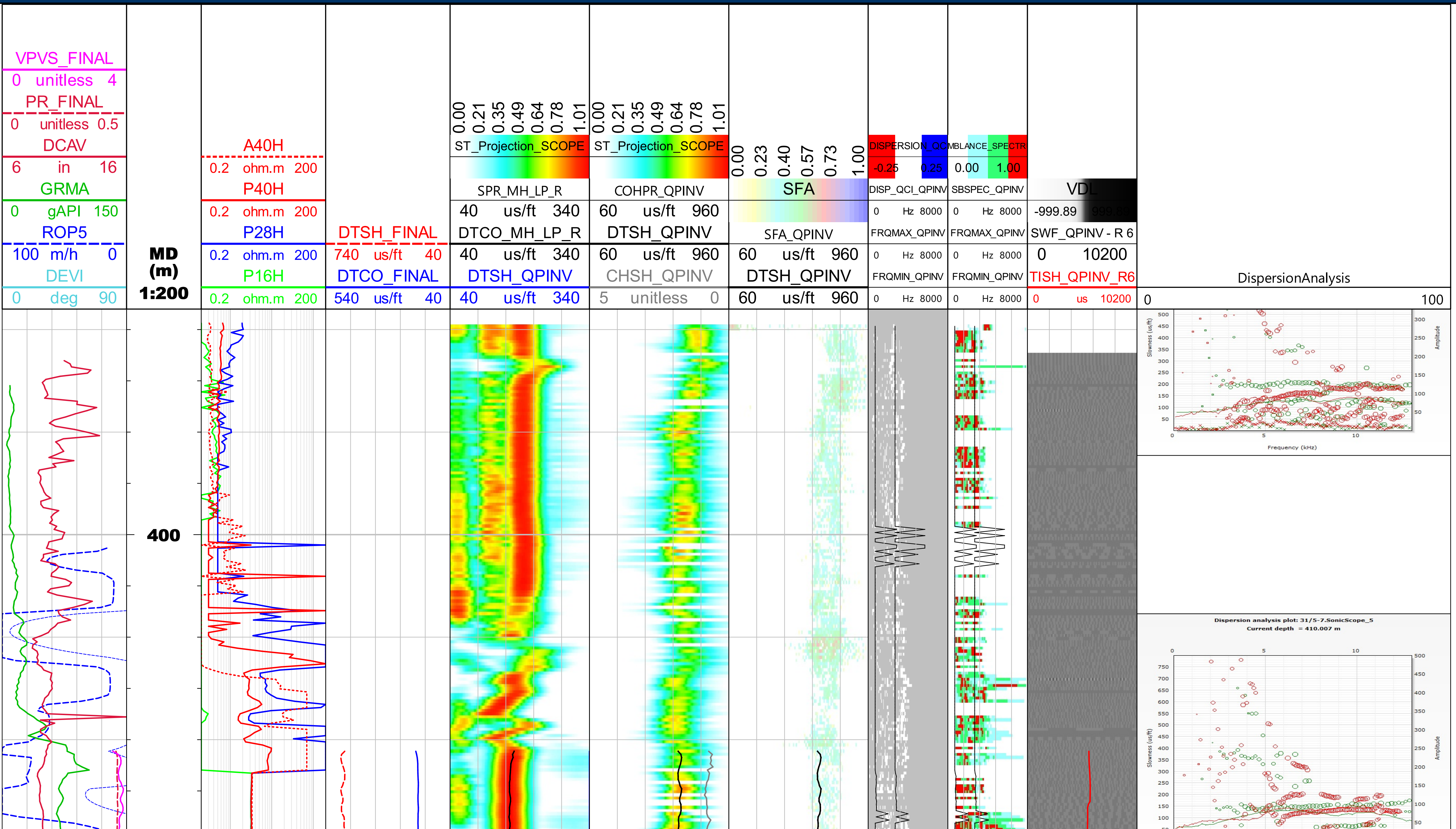
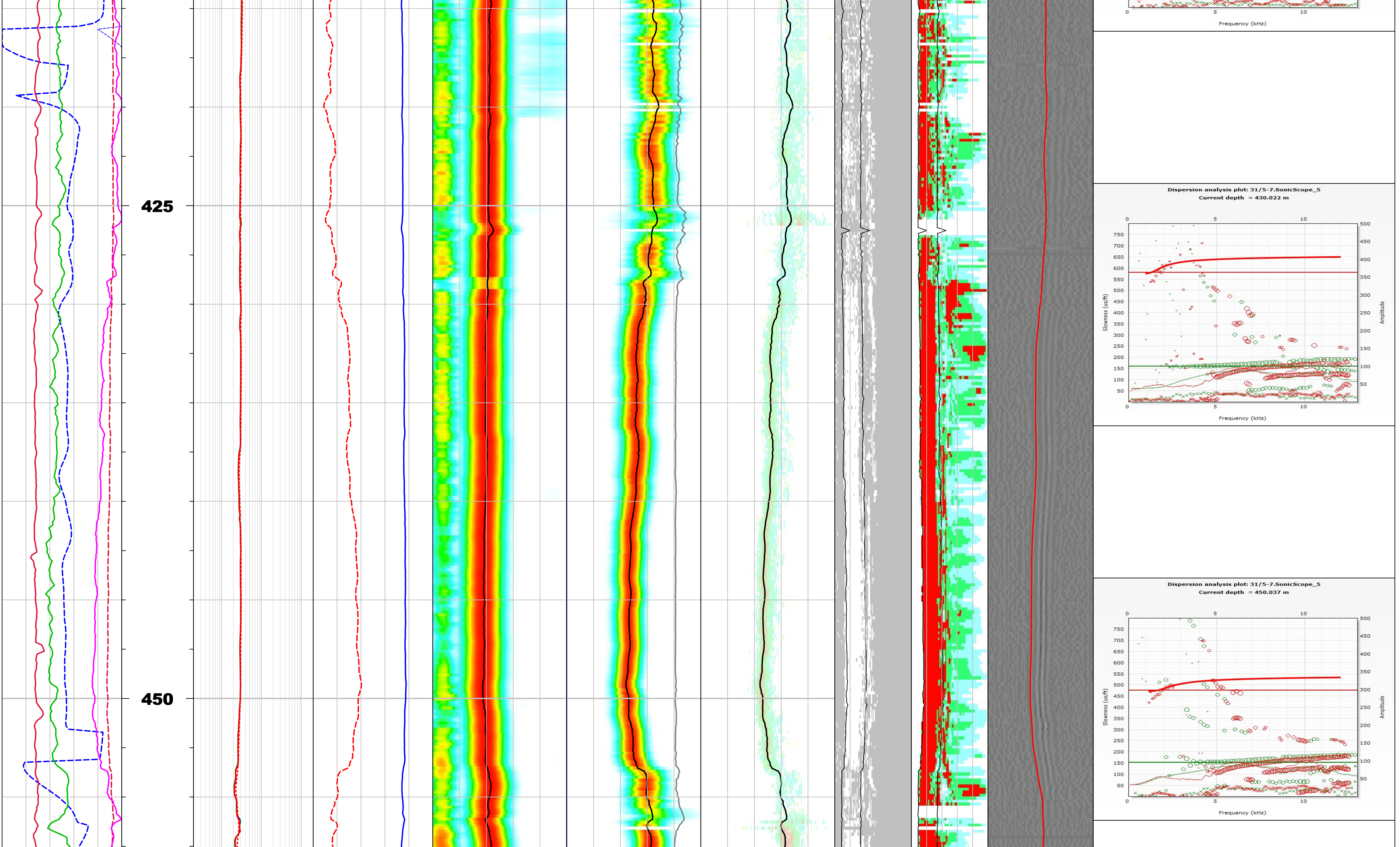
Current depth = 910.387 m
Slowness (us/ft)
Amplitude
Frequency (kHz)
VPVS_FINAL
0 unitless 4
PR_FINAL
0 unitless 0.5
DCAV
6 in 16
GRMA
0 gAPI 150
ROP5
100 m/h 0
DEVI
MD
(m)
A40H
0.2 ohm.m 200
P40H
0.2 ohm.m 200
P28H
0.2 ohm.m 200
P16H
DTSH_FINAL
740 us/ft 40
DTCO_FINAL
0.00 0.16 0.27 0.38 0.49 0.61 0.72 0.83 1.01
ST_Projection_SCOPE
SPR_MH_LP_R
40 us/ft 340
DTCO_MH_LP_R
0.00 0.16 0.27 0.38 0.49 0.61 0.72 0.83 1.01
ST_Projection_SCOPE
SPR_MH_LP_T
40 us/ft 340
DTCO_MH_LP_T
VDL
-73.56 73.56
SWF_MH_LP - R 6
0 5100
TICO_MH_LP_R6
DispersionAnalysis

0 deg 90	1:200	0.2 ohm.m 200	540 us/ft 40	40 us/ft 340	40 us/ft 340	0 us 5100	0 100

DT Multipole Log

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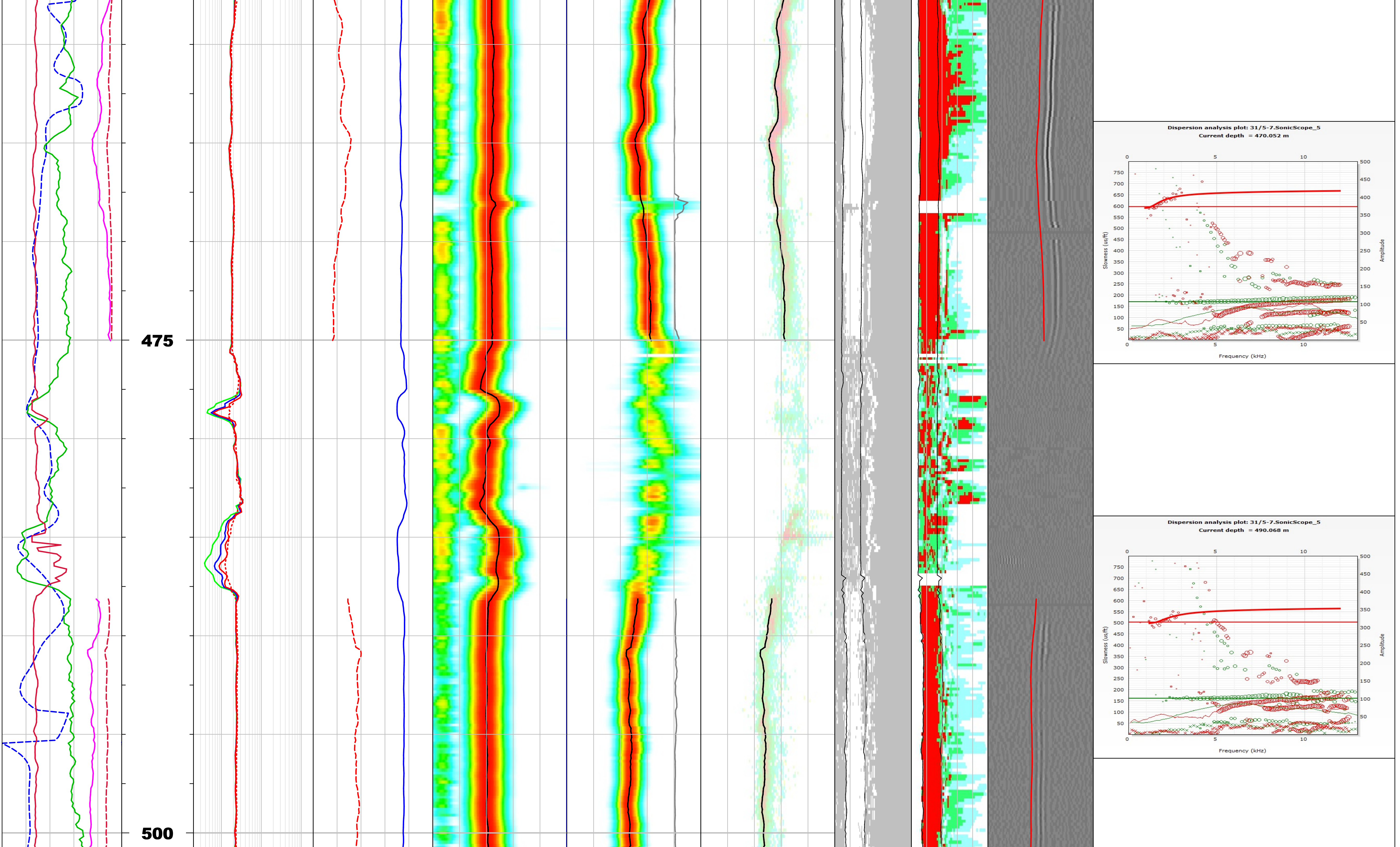
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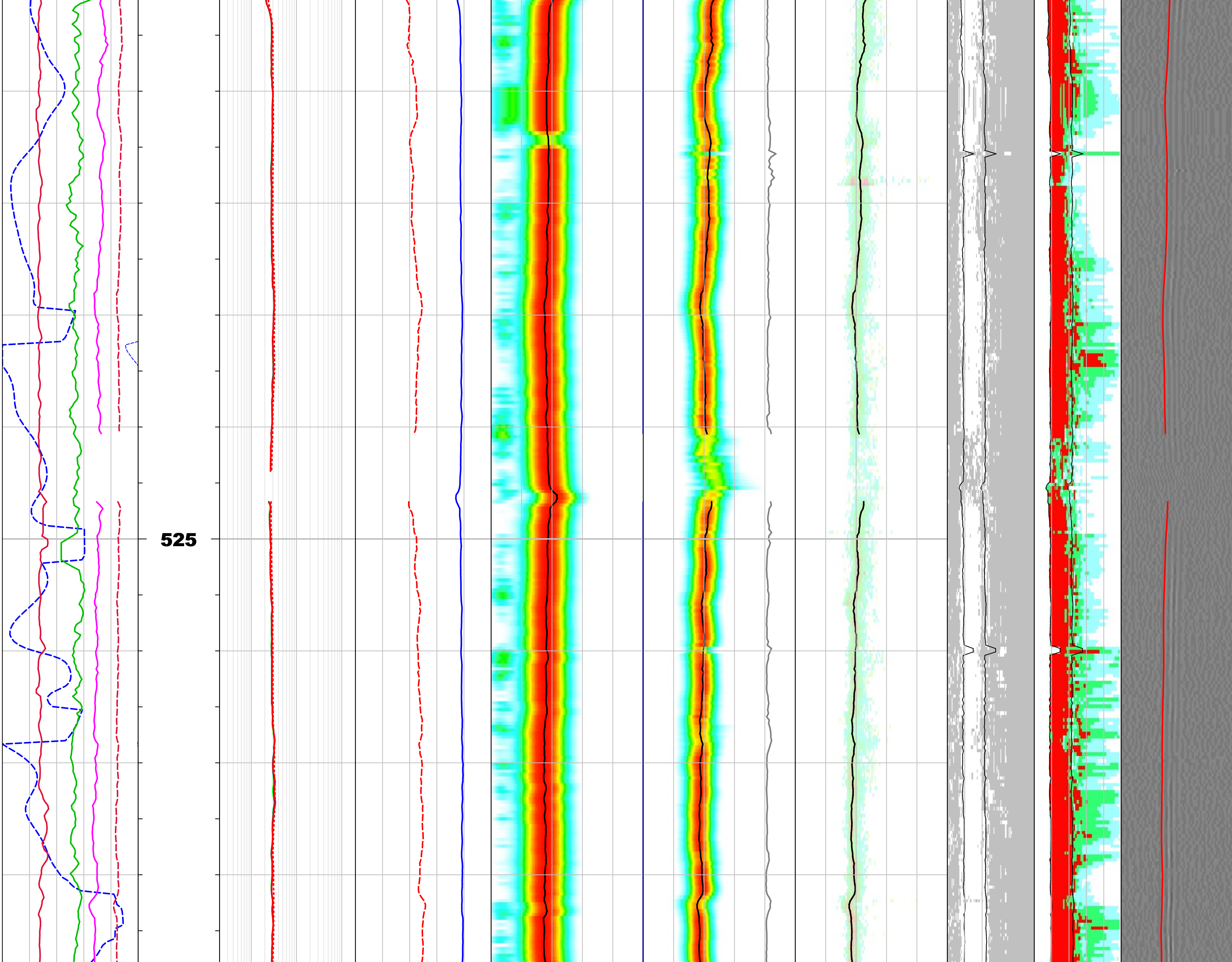
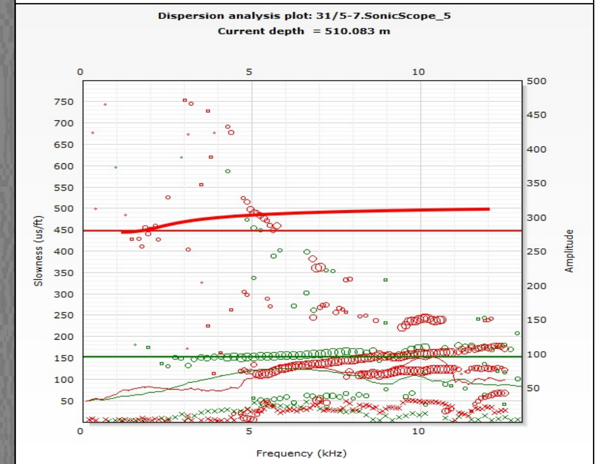

525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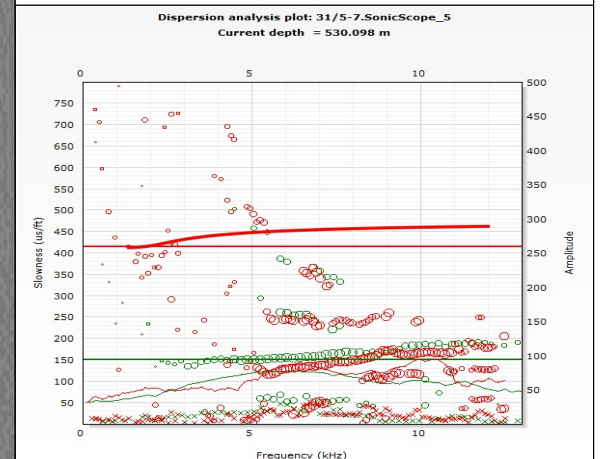
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Current depth = 530.098 m
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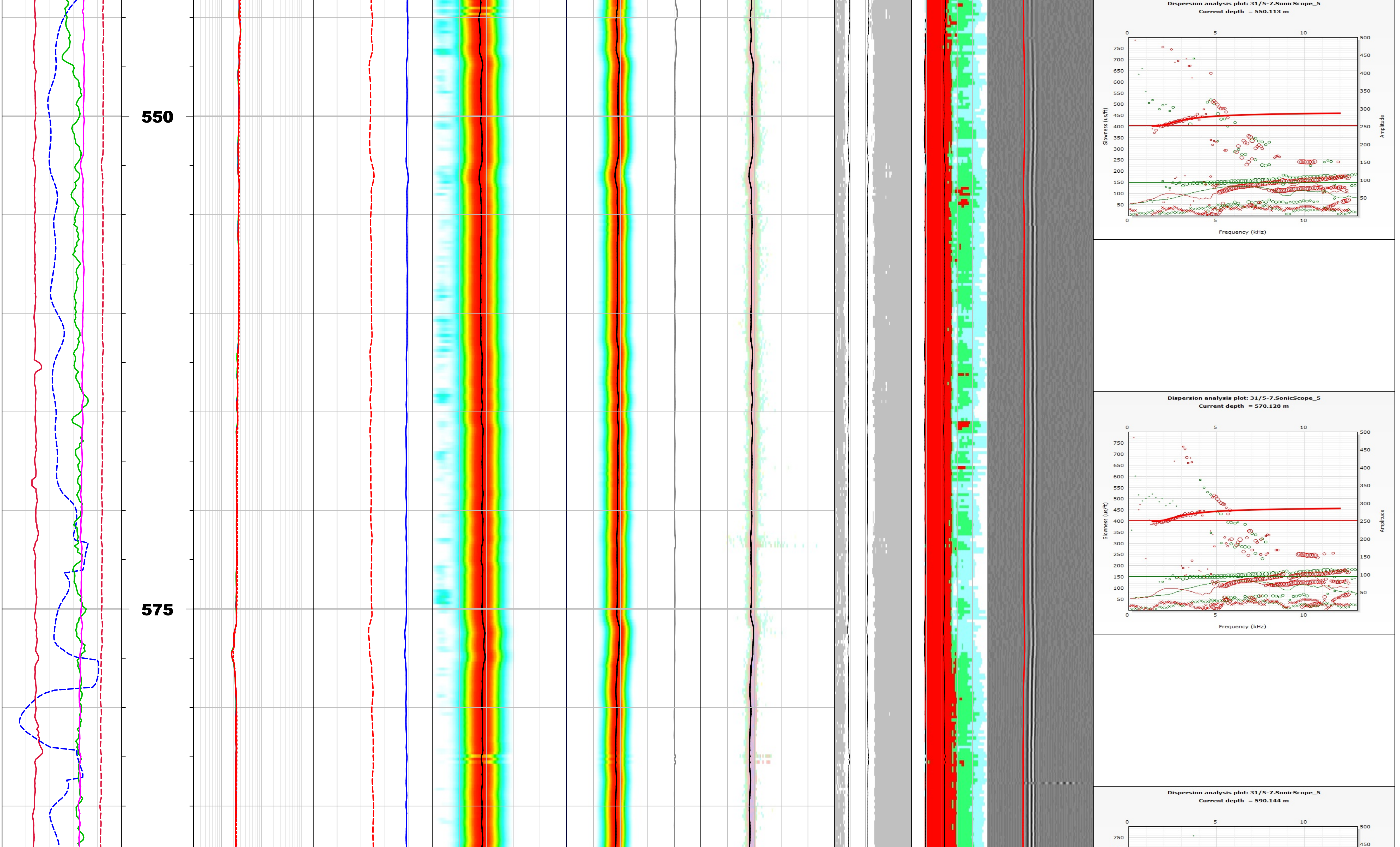

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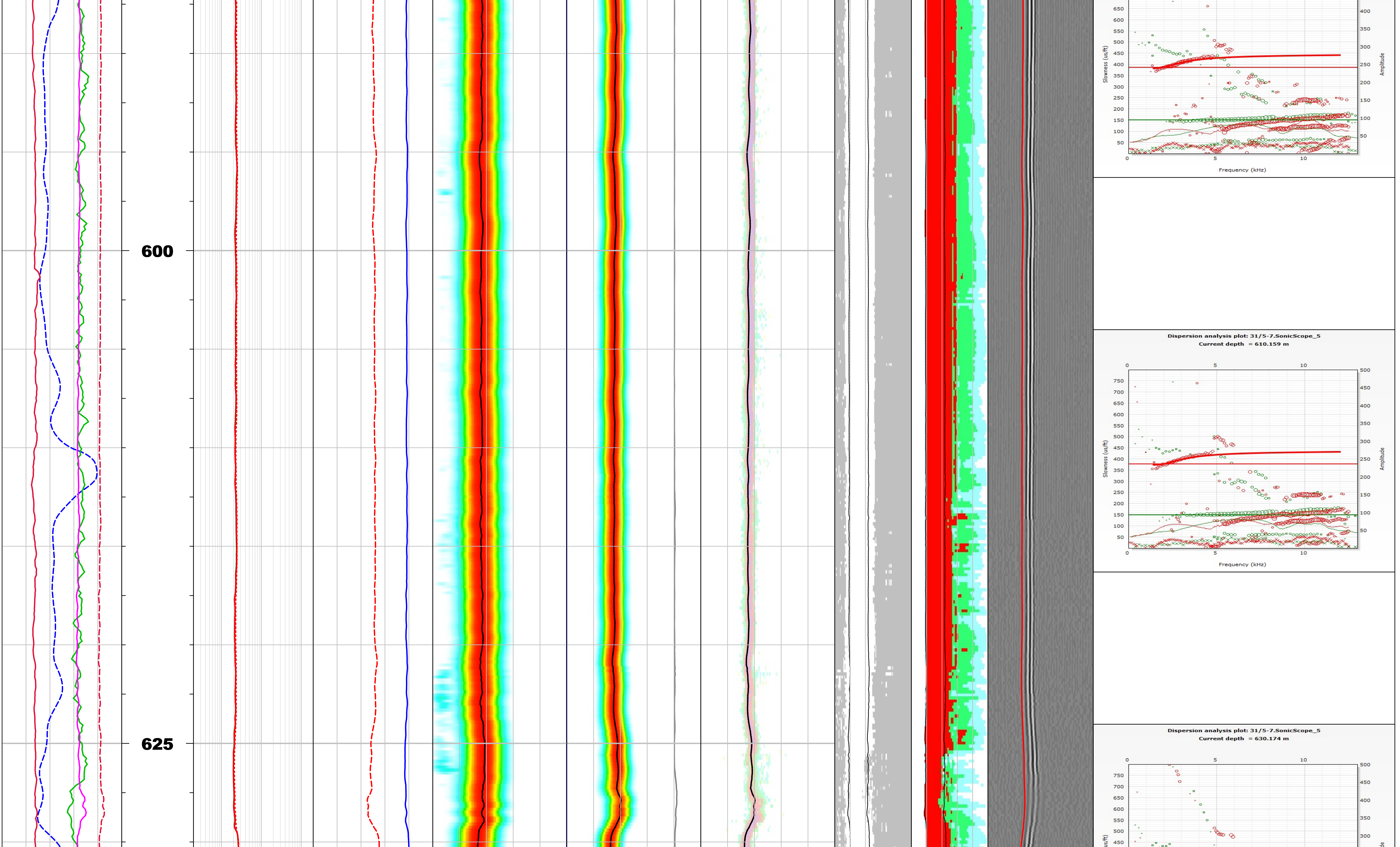
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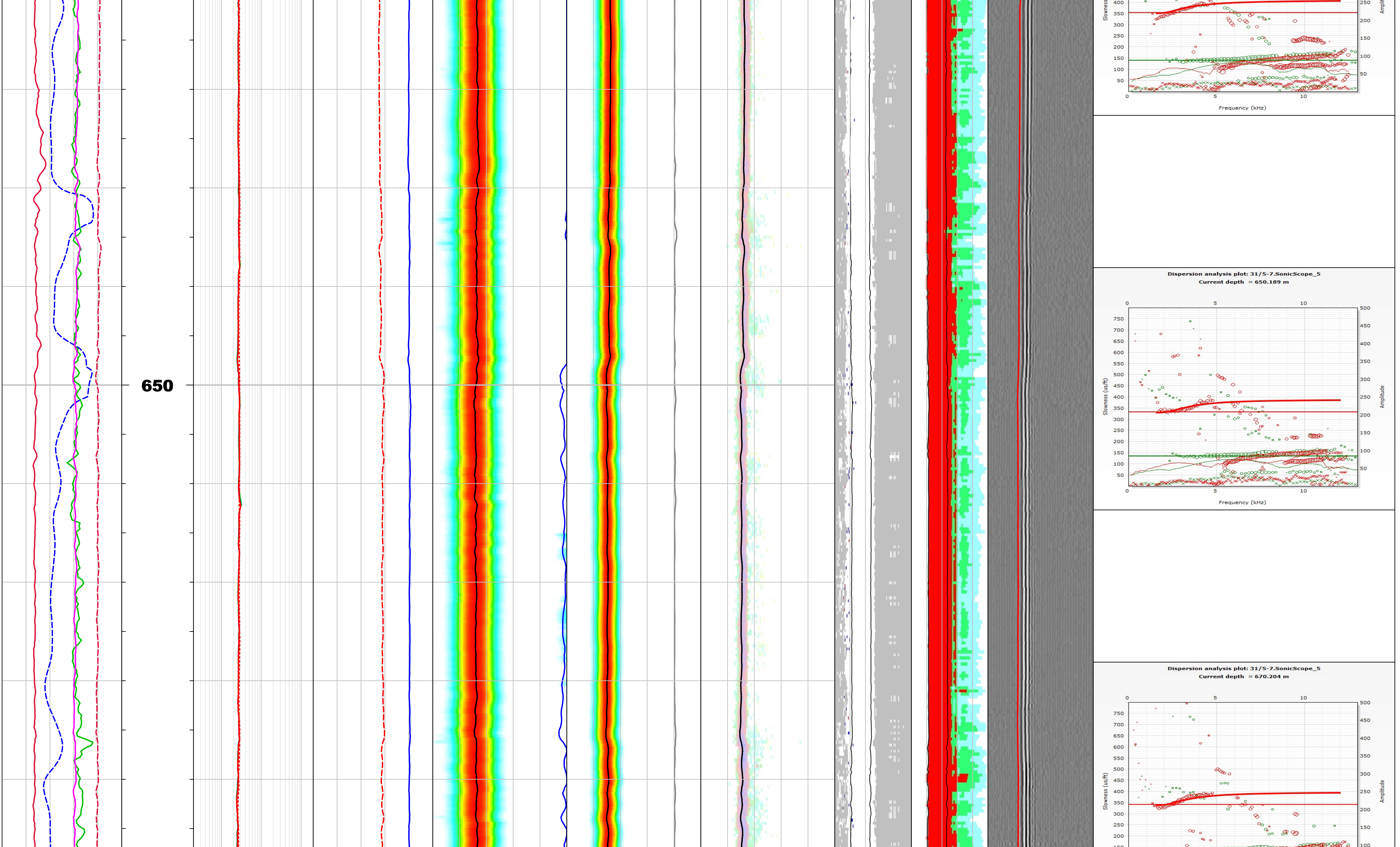
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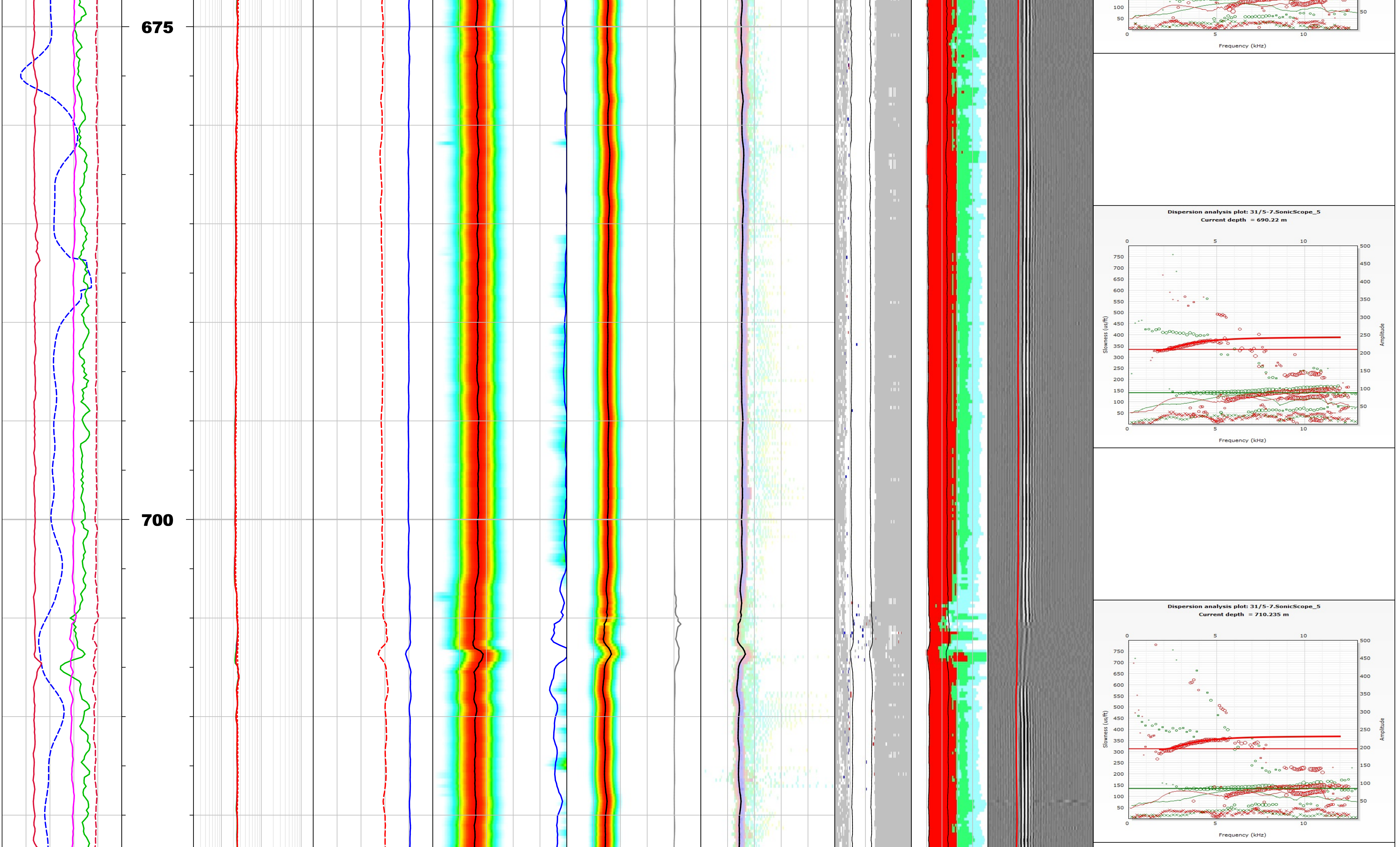
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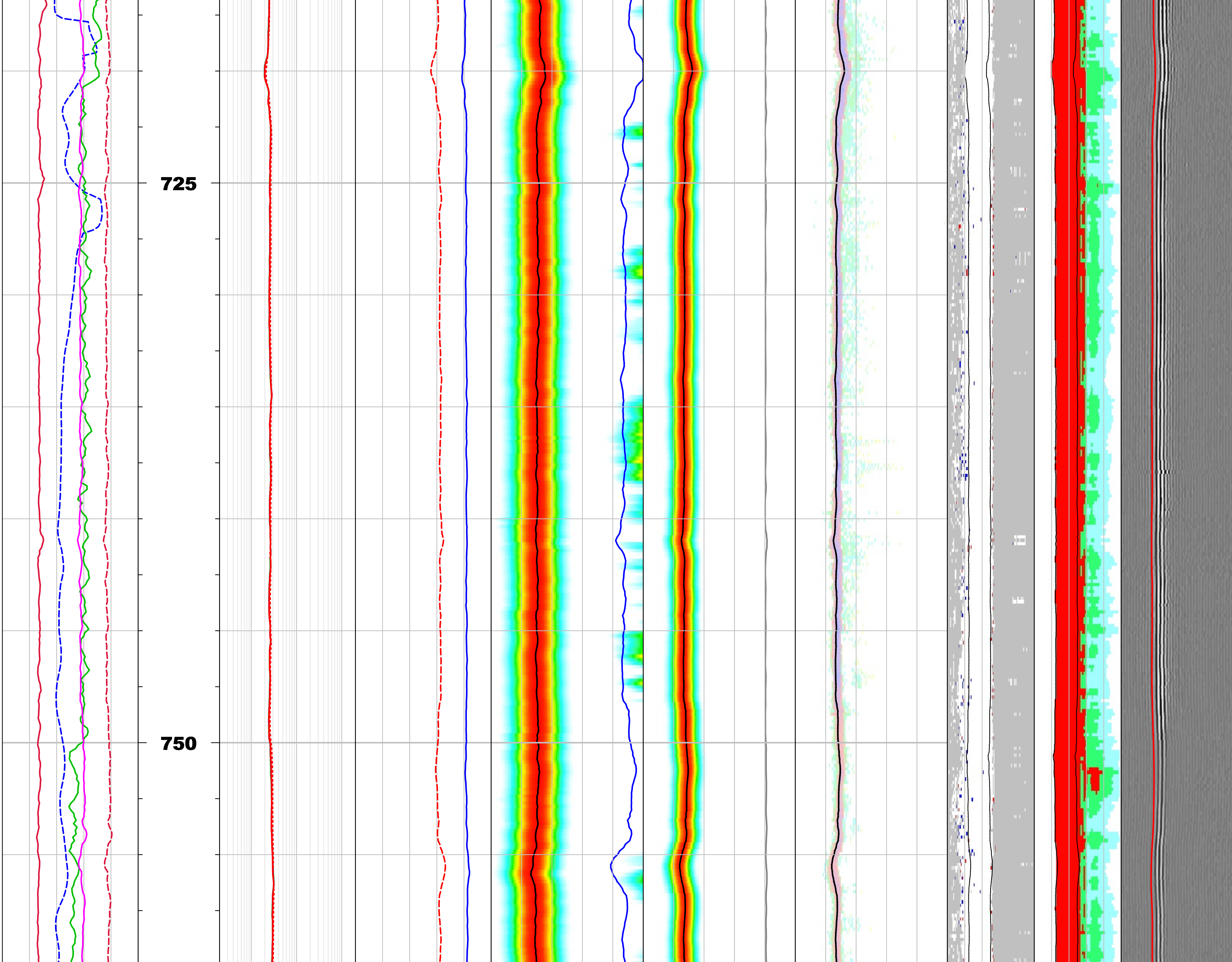
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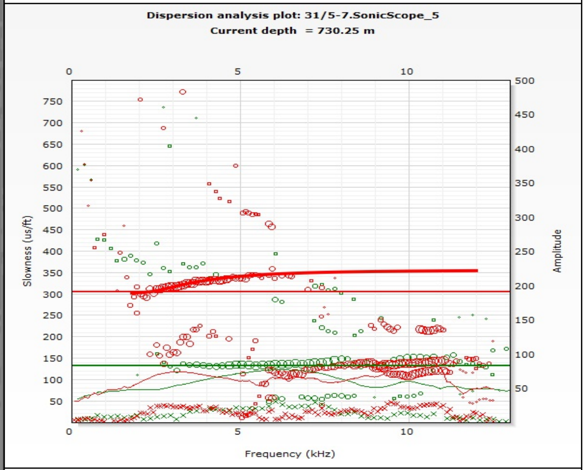
750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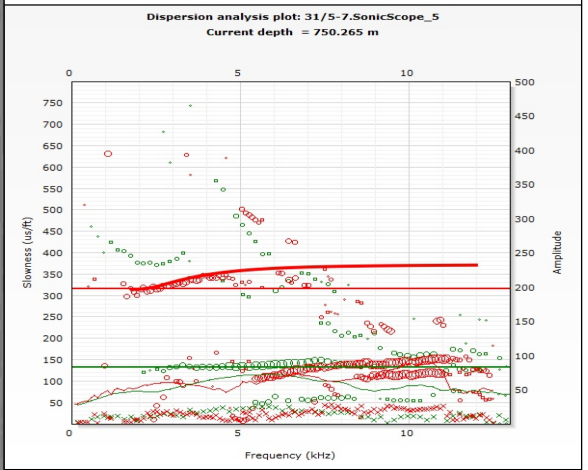
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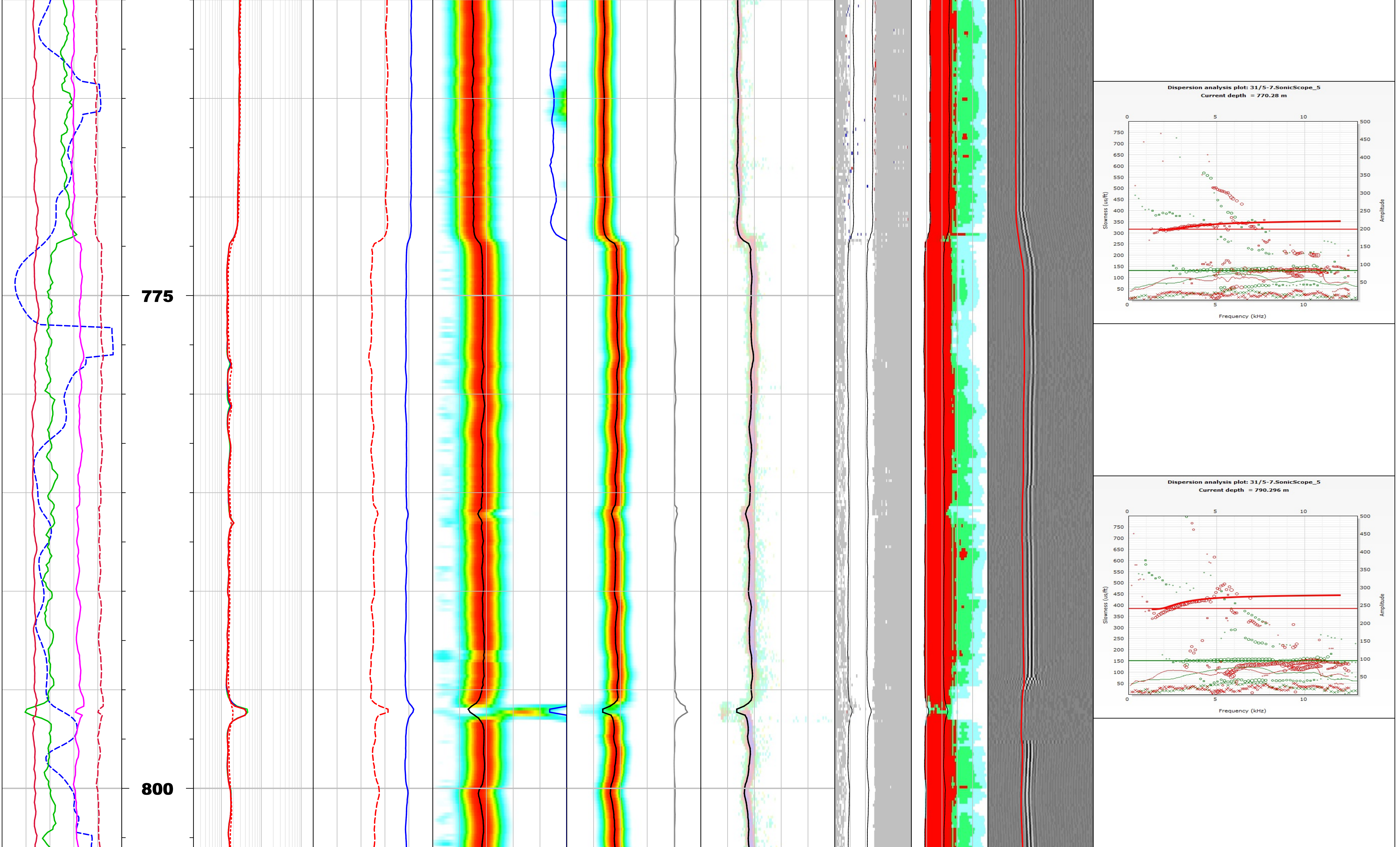
775
800
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Amplitude
Frequency (kHz)
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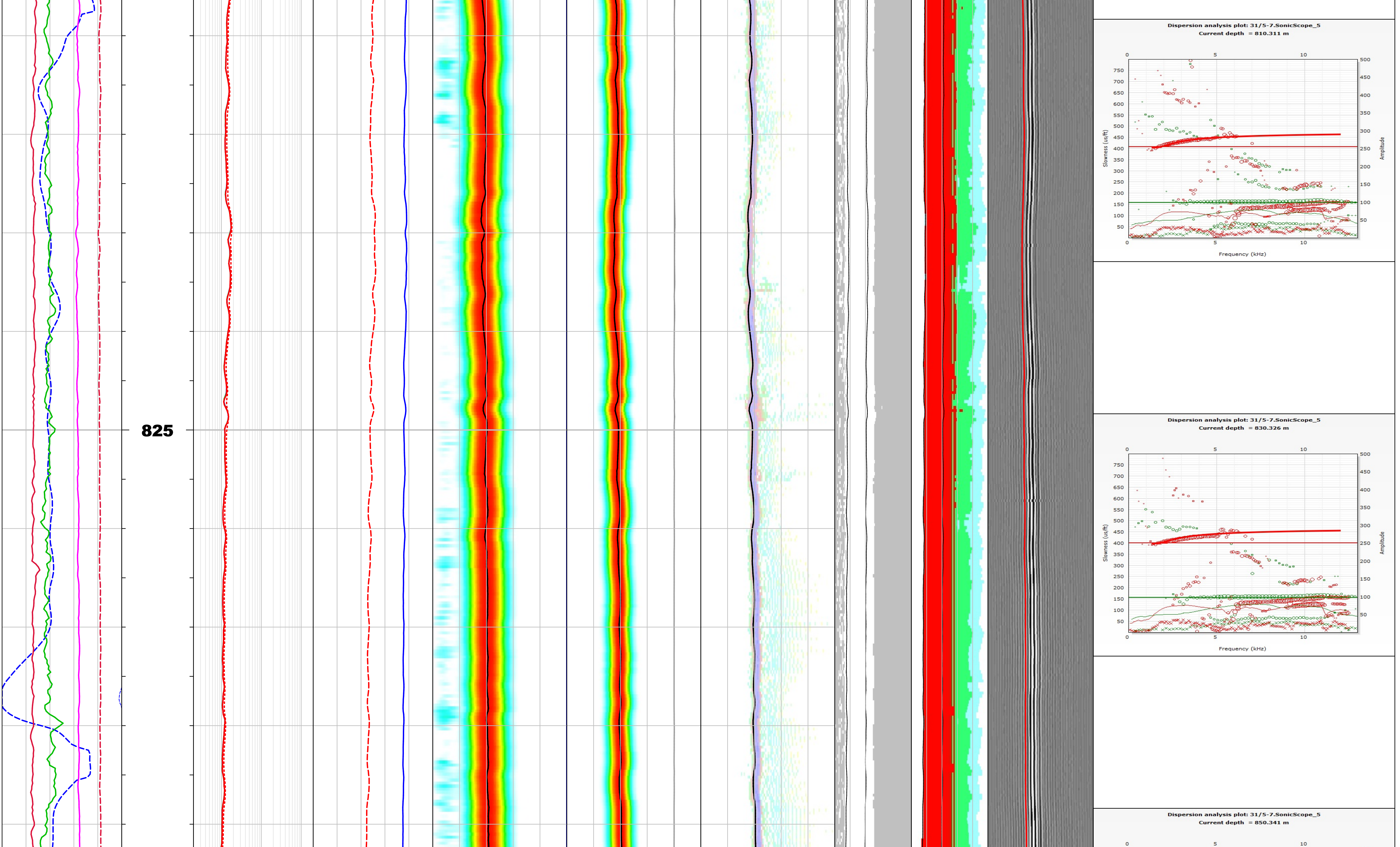
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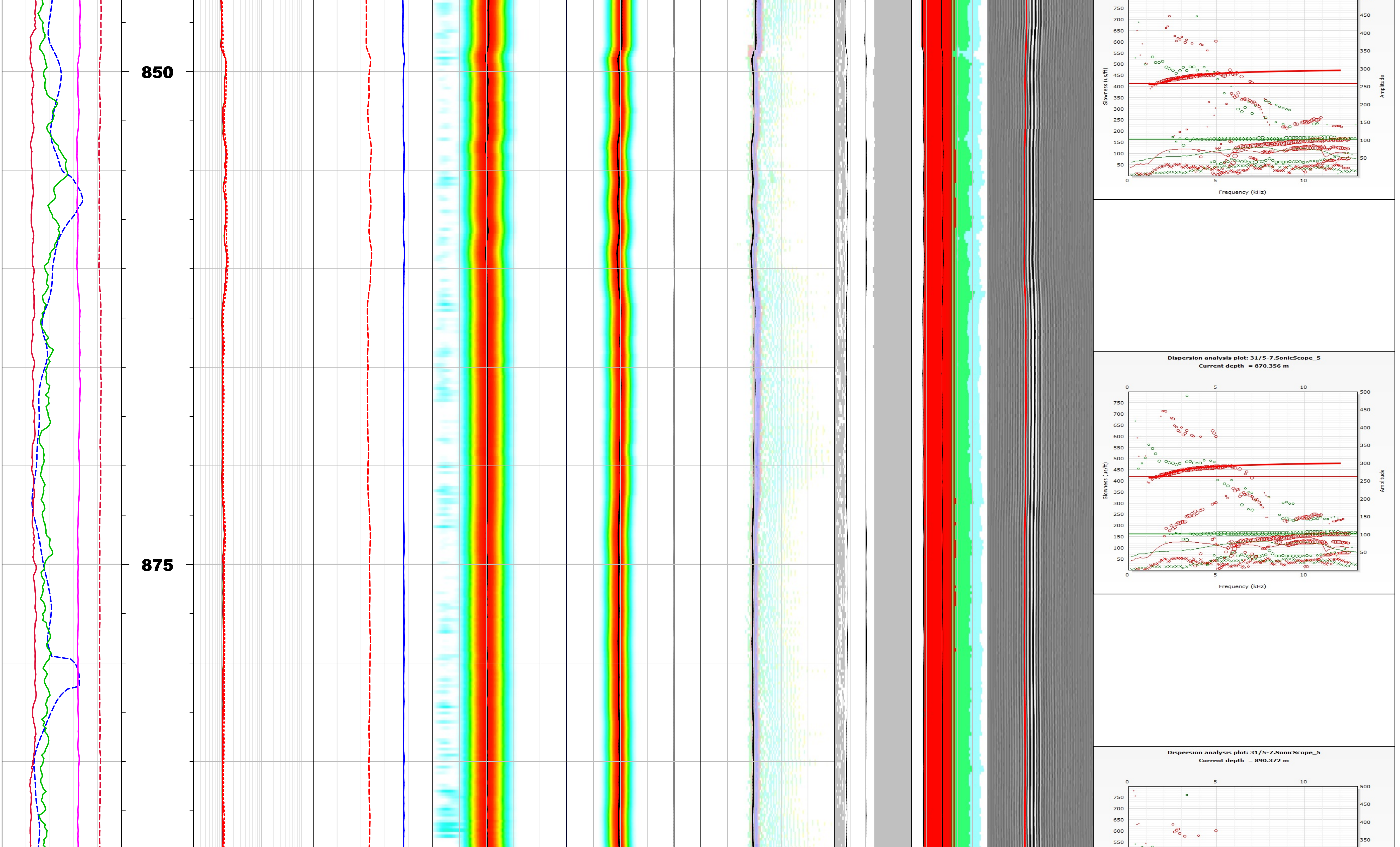
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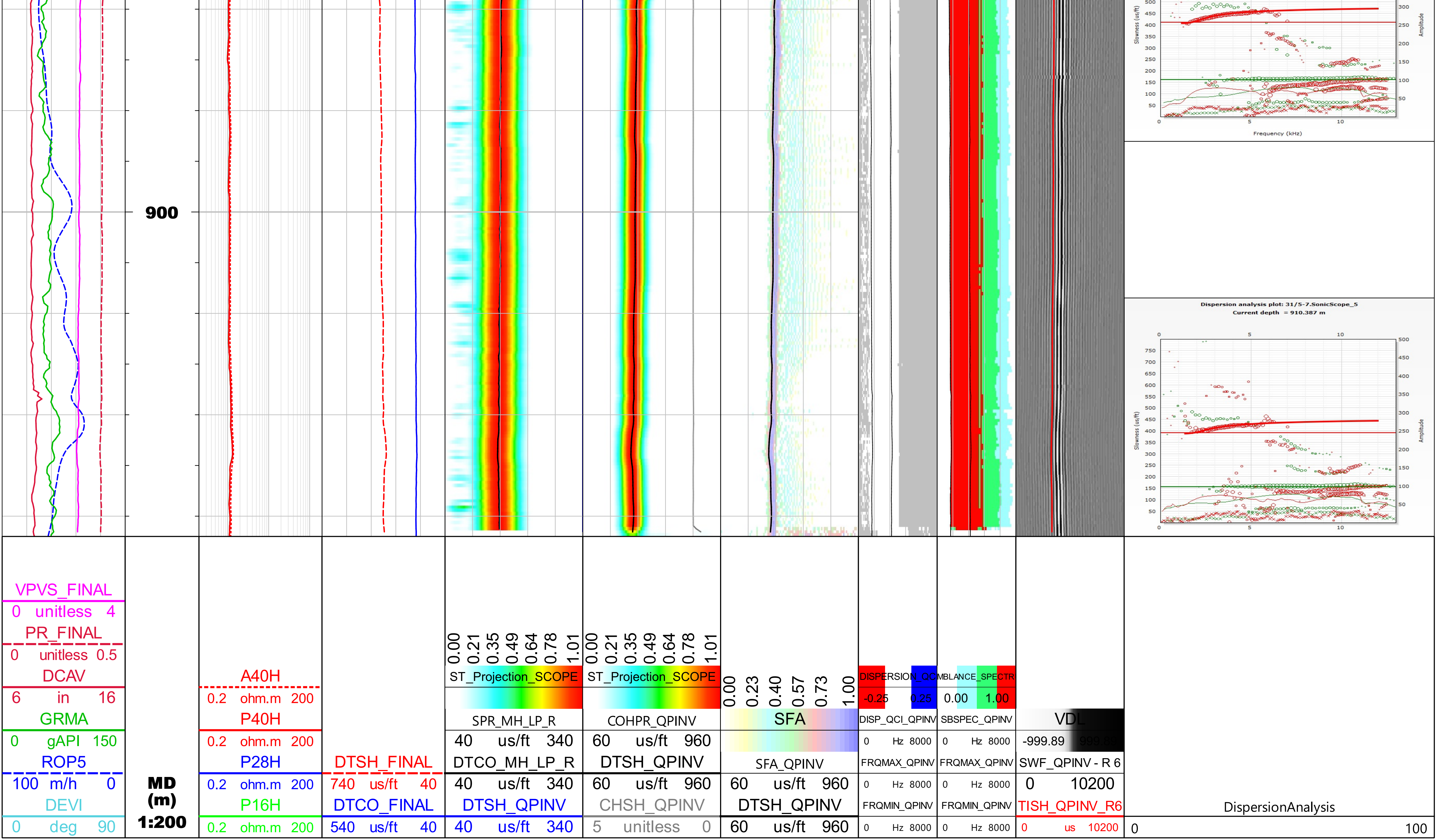
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Frequency (kHz)
Slowness (us/ft)
Amplitude
VPVS_FINAL
0 unitless 4
PR_FINAL
0 unitless 0.5
DCAV
6 in 16
GRMA
0 gAPI 150
ROP5
100 m/h 0
DEVI
0 deg 90
MD
(m)
1:200
A40H
0.2 ohm.m 200
P40H
0.2 ohm.m 200
P28H
0.2 ohm.m 200
P16H
0.2 ohm.m 200
DTSH_FINAL
740 us/ft 40
DTCO_FINAL
540 us/ft 40
0.00 0.21 0.35 0.49 0.64 0.78 1.01
ST_Projection_SCOPE
SPR_MH_LP_R
40 us/ft 340
DTCO_MH_LP_R
40 us/ft 340
DTSH_QPINV
40 us/ft 340
0.00 0.21 0.35 0.49 0.64 0.78 1.01
ST_Projection_SCOPE
COHPR_QPINV
60 us/ft 960
DTSH_QPINV
60 us/ft 960
CHSH_QPINV
5 unitless 0
0.00 0.23 0.40 0.57 0.73 1.00
SFA
SFA_QPINV
60 us/ft 960
DTSH_QPINV
60 us/ft 960
DISPERSION_QC
-0.25 0.25
DISP_QCI_QPINV
0 Hz 8000
FRQMAX_QPINV
0 Hz 8000
FRQMIN_QPINV
0 Hz 8000
MBLANCE_SPECTR
0.00 1.00
SBSPEC_QPINV
0 Hz 8000
FRQMAX_QPINV
0 Hz 8000
FRQMIN_QPINV
0 Hz 8000
VDL
-999.89 999.89
SWF_QPINV - R 6
0 10200
TISH_QPINV_R6
0 us 10200
DispersionAnalysis
0 100

Processing Parameters Summary

Monopole High waveform processing:

Processing parameters summary

Monopole High Leaky-P

Borehole status: Open hole	**Bit size:** 8.5 in	**Caliper:** BS in
Energy threshold: 2 Pa	**Energy threshold option:** All constant energy	**Filter length:** 41
Filter band high: 4500 Hz	**Filter band low:** 2000 Hz	**Filter type:** Static filter
Formation type: Slow	**Average hole size:** 8.5 in	**Leaky-P:** Yes
Dispersion file mode: None	**Receiver/transmitter/DDBHC mode:** Receiver mode	**Mud slowness:** 200 us/ft
Mud density: 1.03 g/cm3	**Mud type:** WBM	**Multishot NRSA:** 7
Search band offset: 1000 us	**Search band width:** 3000 us	**Semblance threshold:** 0.35
Slowness lower limit: 40 us/ft	**Slowness step:** 2 us/ft	**STC stacking:** No
Slowness upper limit: 340 us/ft	**Slowness width:** 20 us/ft	**Compute SFA:** Yes
Max frequency: 12000 Hz	**Min frequency:** 0 Hz	**Model order:** 2
Matching tolerance: 50	**Time lower limit:** 300 us	**Compressional lower limit:** 73.3 us/ft
Compressional upper limit: 165 us/ft	**Shear lower limit:** 112 us/ft	**Shear upper limit:** 772 us/ft
Time step: 300 us	**Time upper limit:** 5100 us	**Time width:** 2250 us
Integration time window: 960 us	**Waveform:** WFA_MHT	**Reference receiver:** 6
Receiver selection: R1;R2;R3;R4;R5;R6;R7;R8;R9;R10;R11;R12		

Quadrupole waveform processing:

Processing parameters summary

Quadrupole inversion

Binning option: Yes	**Bin width:** 1000 Hz	**Caliper:** BS in
Compressional slowness: DTCO_MH_LP_MS7_R_1_RUN6 us/ft	**Window move out (Shear DT):** DTSH_QPINV_MS7_3_RUN2 us/ft	**Leaky-Q filtering:** Yes
Filter band high for filtered waveform: 5000 Hz	**Filter band low for filtered waveform:** 0 Hz	**Filter type:** Auto
Formation type: Slow	**Inversion type:** 2-Parameters	**Monopole high ST Projection:** SPR_MH_LP_MS7_R_1_RUN6 unitless
Mud slowness: 200 us/ft	**Mud density:** 1.03 g/cm3	**Mud type:** WBM
Multishot NRSA: 7	**Bulk density:** RHOB g/cm3	**Max frequency for SFA:** 4000 Hz
Min frequency for SFA: 0 Hz	**Slowness lower limit:** 60 us/ft	**Slowness upper limit:** 740 us/ft
Waveform length: 512	**Max frequency for TKO:** 12000 Hz	**Min frequency for TKO:** 0 Hz
Model order: 3	**Matching tolerance:** 50	**Waveform:** WFA_QPT
Reference receiver: 6	**Receiver selection:** R1;R2;R3;R4;R5;R6;R7;R8;R9;R10;R11;R12	**Processing window start:** 0 us
Processing window width: 6000 us		

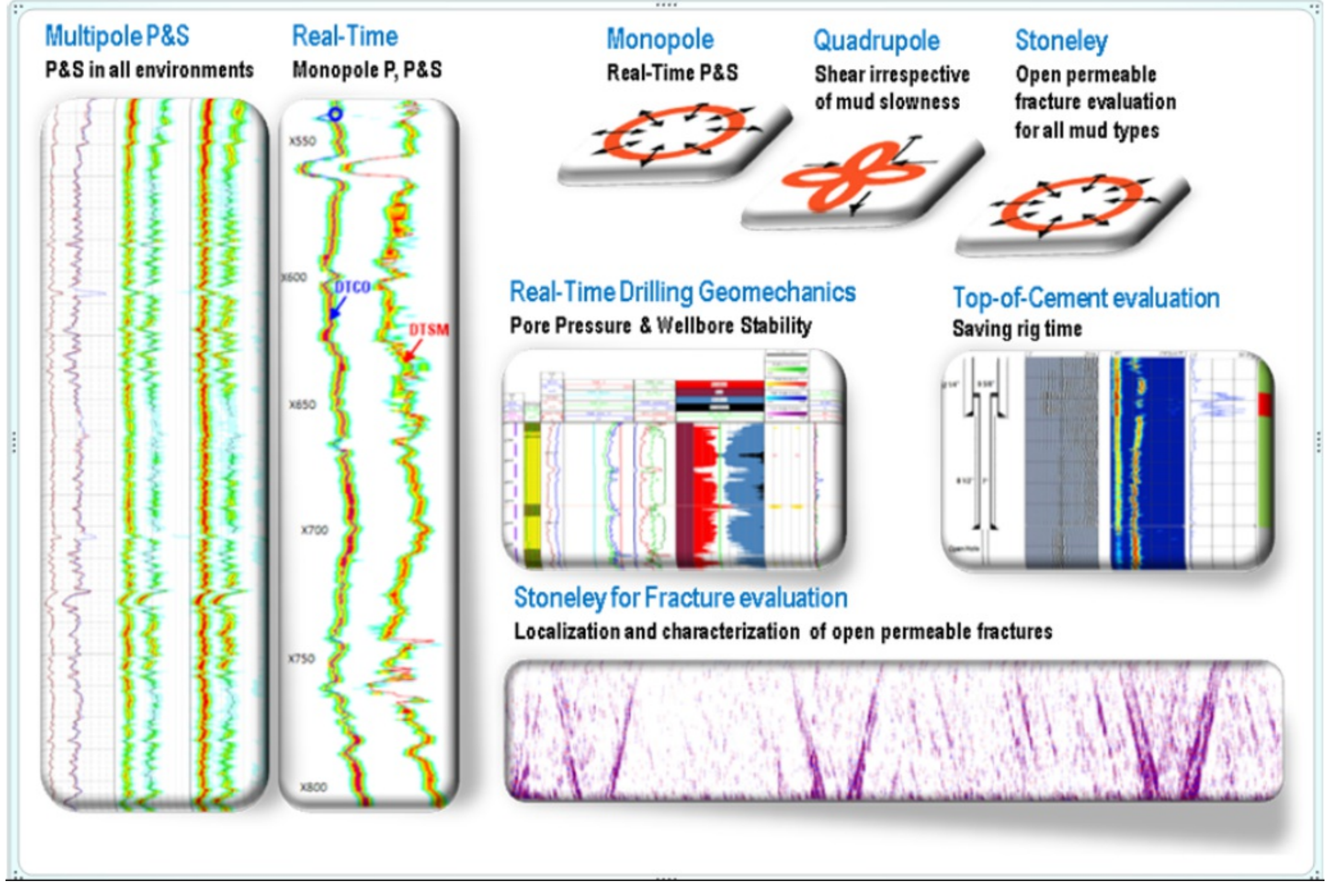
Appendix : SonicScope Service

The SonicScope* multipole sonic-while-drilling service is the latest generation of Schlumberger sonic-while-drilling technologies. It presents a whole new set of capabilities developed through an innovative design, combining multipole measurements for high data quality and robust, advanced solutions. Coupled with extensive modeling, a complete reassessment of strategies for handling and processing acoustic data ensures that the service's tool response is predictable in most environments.

For a wide range of applications, the SonicScope service delivers robust compressional and shear slownesses irrespective of mud speed, combining high-quality monopole and quadrupole measurements. It also supplies enhanced Stoneley data. The sonic information acquired can be used by drillers, geophysicists, geologists, petrophysicists, and reservoir and completion engineers.

SonicScope was developed to fulfill the demand of compressional and shear, both real-time and recorded mode in fast and slow formation*. There is now an increasing push to achieve such requirement in LWD, especially in offshore drilling exploration to development fields.

SonicScope offers the highest sonic while drilling data quality to date on the market and robust output compressional and shear measurements.



The tool is a 'multipole sonic service' including:
1/ Reliable hardware with functionality support further developments
2/ Monopole compressional (P) and shear (S) service for both Real-Time and Recorded Mode
3/ Quadrupole shear in slow formation for recorded mode only
4/ Stoneley products in Recorded Mode (Stoneley for fracture evaluation)
5/ Top of cement evaluation in Recorded Mode

*A fast formation is a formation where the shear slowness (1/velocity) is faster than the mud slowness. A monopole tool can only deliver shear slowness in fast formation. A quadrupole tool can measure a shear slower than the mud speed; i.e. shear in slow formation (same as dipole for wireline).

Appendix : Quadrupole Inversion

The quadrupole (QP) inversion is a model-based inversion to estimate formation shear slowness from waveforms recorded by SonicScope 475, SonicScope 675 and SonicScope 825. The inversion assumes a homogeneous isotropic (HI) formation model including the effect of tool presence on the borehole quadrupole dispersion. The proposed inversion maximizes the semblance that is computed by back-propagating waveforms in the frequency domain using the quadrupole dispersion model (Ref.1).

The borehole quadrupole mode is dispersive, but does not asymptote to formation shear slowness at low frequencies as does the dipole flexural mode. Quadrupole inversion processing can be improved by proper selection of the frequency band to perform the inversion. At low frequencies, the suggested band starts at near the cut-off frequency of the borehole quadrupole mode and stops at near the Airy frequency (peak amplitude frequency). In this frequency band there is higher sensitivity of the quadrupole propagation to formation shear, and also higher signal to noise ratio. This is the band used when the “Auto-select” option is used in the QP inversion processing in Techlog Acoustics Interpretation Suite.

The borehole quadrupole mode is very sensitive to borehole fluid slowness (mud) at high frequencies, especially in fast and intermediate formations. However, mud slowness is not measured directly even though it is needed to properly estimate shear from the quadrupole data. In order to overcome this limitation, a 2-parameter inversion was developed to make the estimated quadrupole shear slowness less affected by undetermined mud slowness. The 2-parameter inversion optimizes the semblance in a 2-D slowness plane, shear slowness and mud slowness. However, this second inversion parameter should not be considered as a good estimate of the true mud slowness, because it includes various environmental effects such as a non-HI formation and errors in the other input parameters. Here we call it an “environmental slowness parameter”.

Reference:

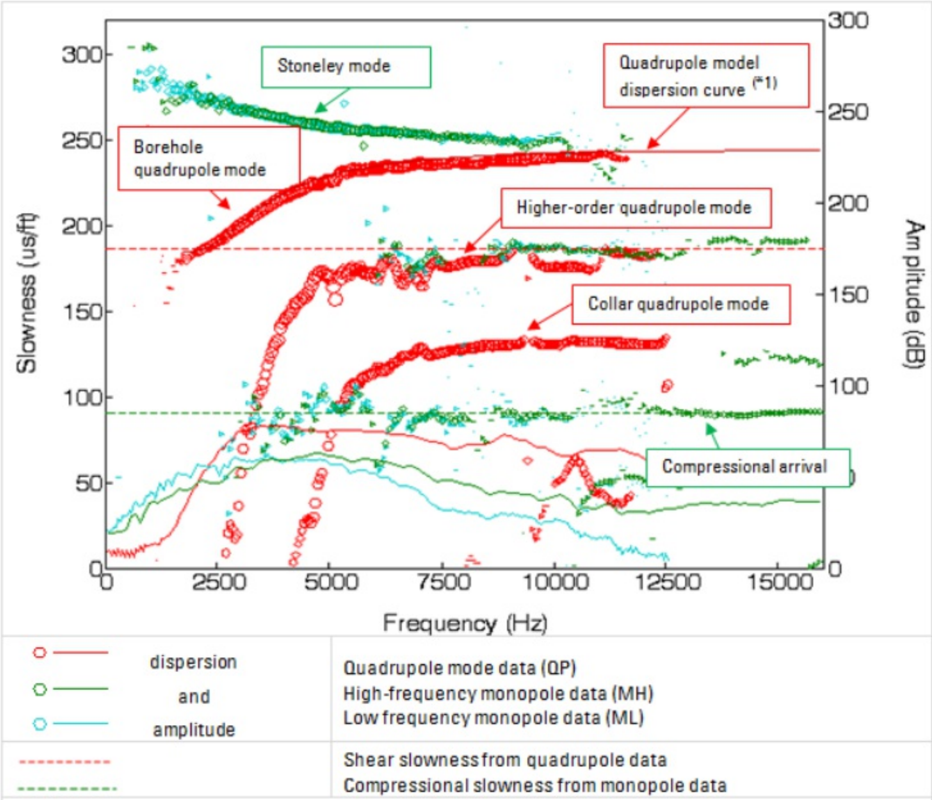
1. Slow Formation Shear from an LWD Tool: Quadrupole Inversion with a Gulf of Mexico Example, Scheibner D, Yoneshima S, Zhang Z, Izuhara W, Wada Y, Wu P, Pampuri F and Pelorosso M, Presented at the SPWLA 51st Annual Logging Symposium, Perth, Australia, 19-23 June, 2010

2. Shear Slowness Estimation by Inversion of LWD Borehole Quadrupole Mode, Zhang Z, Mochida M, Kubota M, Yamamoto H, and Endo T, SEG International Exposition and Annual Meeting, Denver, 2010

Appendix : Dispersion Plot

Slowness dispersion plots in the quadrupole shear quality control plot display processed compressional and shear slowness readings on top of dispersion curves of SonicScope waveform data. In addition, a model dispersion curve of the borehole quadrupole mode based on the formation parameters is calculated and displayed. A good agreement of this model dispersion curve (in solid red line) and the actual dispersion curves of the borehole quadrupole mode (in red circles) in the processing frequency band provides confidence in the accuracy of the processed shear slowness log. The figure explains the legend of lines and symbols used in this plot.

QP model dispersion curves are calculated using formation shear slowness and environmental slowness parameter as well as other formation and borehole parameters input to QP inversion; i.e., compressional slowness, bulk density, borehole size and mud weight. This model dispersion curve considers the effect of the presence of the tool in the borehole and assumes that the formation is homogeneous and isotropic.



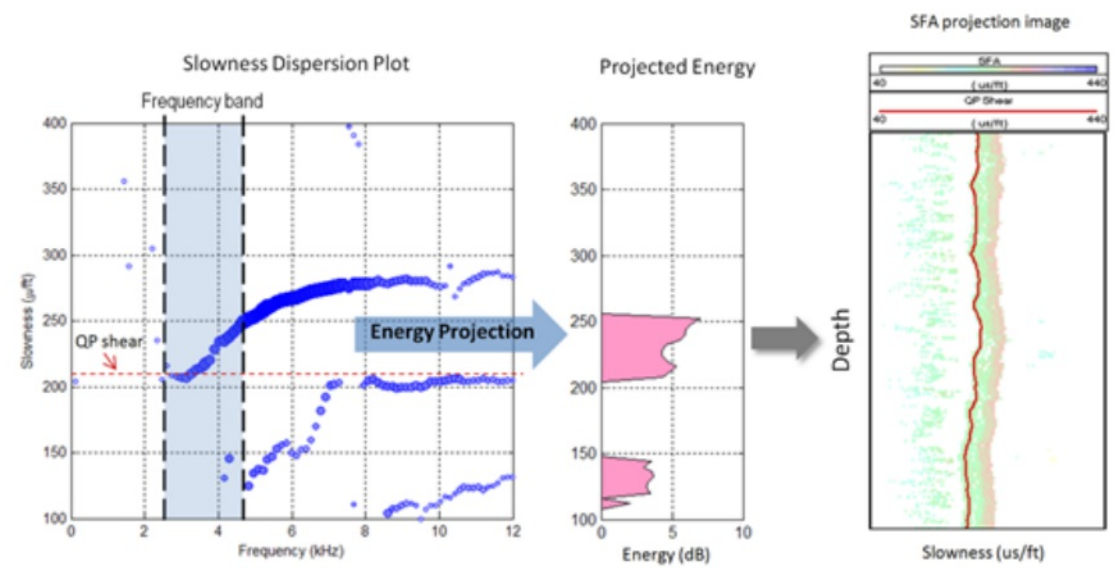
Appendix : Quality Control Indicators

This appendix describes three kinds of quality control indicator image logs that allow assessment of the overall quality of the quadrupole waveform data and processed quadrupole shear slowness log:

- **Slowness-frequency analysis (SFA) image log**
- **Dispersion QC image log**
- **Semblance spectrum image log**

Slowness frequency analysis (SFA) image log

Slowness-frequency analysis image log represents the projection of the dispersions in a certain frequency band onto the slowness axis at each depth as illustrated in the figure below. The color of the SFA image denotes the energy of the propagating waves in the array waveform data. On the SFA image, the shear slowness log is overlaid (red curve in figure below), and it should lie near the left edge of the SFA image, indicating that the shear slowness log is at the lower (faster) edge of the quadrupole dispersion curve. If this is not the case, it is an indicator to check the full dispersion plot at those depths to understand why. The shear slowness log may be good and the image to the left of the shear log may come from other expected arrivals or noise, or perhaps the processing parameters need adjustment, or maybe there is eccentering, alteration, other modes, etc. In any case, viewing the full dispersion display at particular depths usually makes the reason.



Reference:

Slowness-frequency projection logs: A new QC method for accurate sonic slowness evaluation: Plona T, Kane M, Alford J, Endo T, Walsh J and Murray D, 46th SPWLA Annual Logging Symposium, New Orleans, 26 - 29 June, 2005

Dispersion QC image log

The dispersion QC image compares, at each frequency, the data dispersion to the model dispersion from the inversion. It is a QC indicator to see how well the model from QP inversion results fit the data.

- White indicates that the difference between the data and model is within 8%.
- Red indicates that the data is faster than the model by more than 8%.
- Blue indicates the data is slower than the model by more than 8%.
- Grey indicates that there are no data dispersion values within 25%.

Therefore, if red or blue colours extend over the frequency band at certain depths, it usually means that the data dispersion is not well defined over a wide band, the formation type deviates significantly from the HI model, or the QP shear is not determined appropriately. In most cases, checking the dispersion curves allows determination of the real causes. If the image is mostly white in a depth interval, it means that the model from the inversion result and the data are in reasonable agreement.

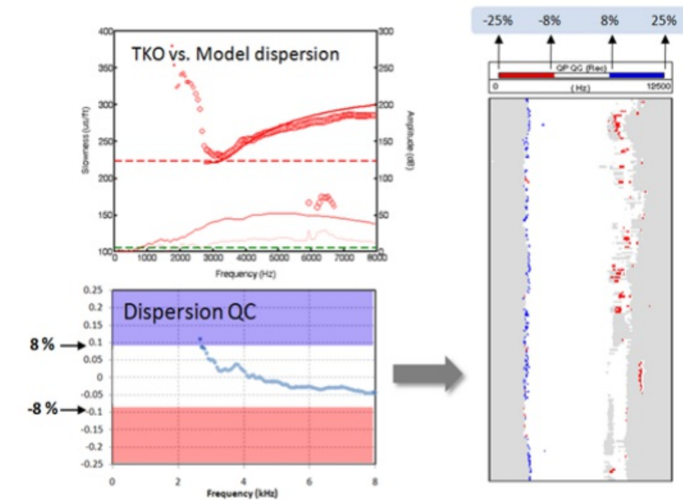
Semblance spectrum image log

The semblance spectrum image shows whether the semblance is high (red), medium (green), or low (grey or white) at each frequency at each depth. This image log shows the width of the band of frequencies over which the quadrupole data is coherent. High semblance in the frequency band of the QP inversion provides confidence in the shear slowness result.

Dispersion QC image

- Relative difference between data dispersion and model dispersion
- Indicator of dispersion match from inversion

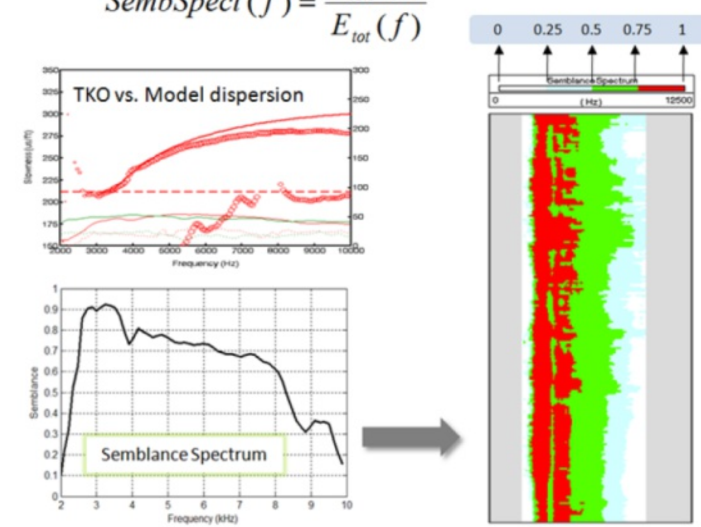
$$Dispersion\ \ QC\,(f) = \frac{Data\ (f) - Model\ \ (f)}{Data\ (f)}$$



Semblance Spectrum

- Calculate semblance at each frequency
- Indicator of coherent energy against total energy

$$SembSpect\,(f) = \frac{E_{coh}\,(f)}{E_{tot}\,(f)}$$



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SonicScope Processing

Techlog - SIS Processed Results